Module description Catalog Program in Mathematics and Computer Science

FACULTY OF MATHEMATICS AND COMPUTER SCIENCE
UNIVERSITY OF SCIENCE
VIETNAM NATIONAL UNIVERISTY HO CHI MINH CITY

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I. GENERAL EDUCATION

Basic Economics - BAA00005

Module designation	Basic Economics
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Le Nhan My
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	Total workload: 90 hours. Contact hours (lecture): 30 hours. Private study including examination preparation, specified in hours: 60 hours.
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Knowledge: Forming and developing (one step) capacity to collect information, skills to synthesize and systematize issues in an overall relationship; skills to compare, analyze, comment, and evaluate micro-economic issues. Grasp the basic content of Microeconomics - a part of economics. Specifically: • Understand the theory of economic choice, the influence of the law of scarcity, and economic models on economic choice. • Understand the theory of supply and demand. • Understand the theory of consumer behavior. • Understand the theory of producer behavior. • Understand the theory of competition and monopoly. • Understand the theory of factor markets.

	• Understand the theory of the role of government. • Understand the analysis of the influence of factors on the balance of the market, in terms of skills. • Having the ability to apply the knowledge learned to study the nature of economic phenomena, the laws, and trends of the phenomena, and the laws of the market economy. • Ability to apply the knowledge learned in the study of macroeconomics, development economics, and several other economic subjects. Skills: • Forming and developing (one step) capacity to collect information, skills to synthesize and systematize issues in an overall relationship; skills to compare, analyze, comment, and evaluate micro-economic issues. • Develop reasoning and public speaking skills. Attitude: • Trying to be righteous in recognizing and evaluating the lines, policies, and laws of the State of Vietnam in the development of the market economy with the state's regulation. Other Objectives: • Through presentations and problem-solving. Forming and developing collaboration and teamwork skills. • Develop skills of creative thinking, discovery, and discovery. • Cultivate and develop assessment and self-assessment capacity. • Develop public speaking and commenting skills.
Content	The course presents some basic problems of economics; principles of economics, supply and demand patterns and market equilibrium; theory of consumer behavior and business behavior; types of markets; aggregate supply, aggregate demand, and measure national output. The module includes the following chapters: Chapter 1: Economics and Fundamental Issues Chapter 2: Markets, Supply and Demand, and Market Equilibrium

	Chapter 3: Theory of Consumer Behavior Chapter 4: Production Cost Theory and Profit Maximization Chapter 5: National Output Measurement, Inflation, and Unemployment
Examination forms	Homework: 20%. Midterm exam: 20%. Final exam: 60%.
Study and examination requirements	Regulations on time, attendance, and discipline in the course: attend class on time and at least 70% of the sessions (only to be absent for a maximum of 30%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have all test scores, lively discussions, constructive and serious statements in class.
Reading list	None

Earth Science - GEO00002

Module designation	Earth science
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at Faculty of Geology, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	Total: 60 hours. Contact hours: 30 lectures hours (in class) Private study: 30 hours (self-study)
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Earth Science introduces general knowledge about the Earth related to the rights of the Earth, inside and outside the Earth including position and operation of the earth in space;

	composition and structure of the atmosphere, wind, and weather, climate, climate change; distribution of freshwater in hydrosphere, sea and ocean behaviour, El Nino-La Nina phenomena: Geosphere: composition and structure of the earth's crust, weathering, erosion-accumulation, landslides, the internal structure of the earth, earthquakes, volcanoes, plate tectonic activities; learn about the history of the earth through the record of fossil remains. Knowledge of Earth science is a necessary basis for understanding the natural environment of the Earth.
Content	1. Presentation on deformations of the earth's crust and earthquake, volcanic activities and membrane tectonic mechanism. 2. Learn about Earth's history through fossil and stratigraphic records. 3. Apply this knowledge to explain some issues in the main profession. 4. Skilled in group discussion, presentation and criticism Attitude, diligence: enthusiasm, honesty in learning; Serious and honest in checking.
Examination forms	Writing (Midterm: 30%, Final exam: 70%)
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Earth Science*, DANIELSON, E.W., DENECKE. EJ. Ir. 1986. 2. *Foundations of Earth Science*, Lutgens Frederick K. Tarbuck Edward, 1, 1997. 3. *Earth Science Textbook*, LUU DUC HAI, TRAN NGHI. 2008.

English 1 - BAA00011

Module designation	English 1
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at Foreign language centre of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory

Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading, and Writing. Here are the objectives in detail: Learners will be able to understand and use vocabulary in various topics such as leisure activities, important life events, emotion, attitude, physical appearance description, travel plans, presenting dreams, countries, people, and languages. Learners can understand and use grammar structures at the pre-intermediate level such as basic tenses and other related matters. Learners will be able to choose the answer that best describes the given picture, choose the correct response to the questions, and understand dialogues and short monologues. Learners will be able to pronounce single words, word clusters and sentences, describe a given picture, and build basic communications in daily life. Learners will be able to comprehend 300-500 words passage of familiar topics and gain more knowledge of different cultures around the world. Learners can write essays about familiar topics related to daily life, learning activities, entertainment, events, etc.

Content	This course is designed for non-English major students at the University of Science - Vietnam National University - Hochiminh City, using the first eight modules in the book New Cutting Edge (Pre-intermediate). These modules cover vocabulary, grammar, reading, listening, speaking, and writing in a wide range of topics such as leisure activities, important life events, feelings and emotions, attitudes, physical appearance descriptions, travel plans, presenting dreams, countries, people, and languages. Students need to complete various tasks, including presentations, debates, role-plays, assignments, tests and so on.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: workbook*. Harlow: Pearson Education.

English 2 - BAA00012

Module designation	English 2
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at Foreign Language Centre of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)

Required and recommended prerequisites for joining the module	Required prerequisites: • English 1 (BAA00011)
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners will be able to understand and use vocabulary in various topics such as everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, finance- related issues. Learners can understand and use grammar structures in pre-intermediate level such as basic tenses and more complex grammatical structures including conditional sentences, passive, and verb patterns. Learners will be able to choose the correct response for the questions and understand dialogues and short monologues. Learners will be able to pronounce words, generate short conversations, discuss real-life familiar topics, understand, and quickly respond to generated questions, and improve basic communication skills in daily life. Learners will be able to comprehend 500 - 700 words passage of familiar topics and gain more knowledge of different cultures around the world. Learners can write appropriate responses to written requests or complaints in business and social contexts, applying theories into real life practice.
Content	This course is designed for non-English major students at the University of Science - Vietnam National University - Ho Chi Minh City, using seven modules (modules 09-15) in the book New Cutting Edge (Pre-intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics such as everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, and finance-related issues. Students need to complete various tasks, including presentations, debates, role-plays, assignments, tests and so on.

Examination forms	Writing (Midterm: 30%, Final exam: 70%)
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: workbook*. Harlow: Pearson Education.

English 3 - BAA00013

Module designation	English 3
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Foreign Language Centre of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours. Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • English 2 (BAA00012)
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners will be able to understand and use vocabulary in various topics such as leisure activities, important life

	events, emotion, attitude, physical appearance description, travel plans, dreams, countries, people, and languages. Learners can understand and use new languages in a natural, communicative way. Learners will be able to present their opinions about some social and cultural issues and understand dialogues and talks. Learners will be able to comprehend 500-700 words passages of familiar topics and gain more knowledge of different cultures around the world. Learners can write paragraphs about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English majors at the University of Science - Vietnam National University - Ho Chi Minh City, using the first six modules in the book New Cutting Edge (Intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics, namely leisure activities, important life events, feelings and emotions, attitudes, physical appearance descriptions, travel plans, presenting dreams, countries, people, and languages. Students need to complete various tasks, including presentations, debates, role-plays, doing homework, tests and so on.
Examination forms	Writing (Midterm: 30%, Final exam: 70%)
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: workbook*. Harlow: Pearson Education. 3. Materials prepared by the lecturer.

English 4 - BAA00014

Module designation	English 4
Semester(s) in which the module is taught	4th semester

Person responsible for the module	Lecturers at Foreign Language Centre of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours. Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • English 3 (BAA00013)
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their intermediate knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners can understand and use the language needed in more complex real-life situations in a natural, communicative way. Learners will be able to express their own ideas in interviews, mini-talks, problem-solving and storytelling. Learners will be able to comprehend 700-1000 words passages of up-to-date topics of international interest and learn more about the world and other cultures. Learners can write essays about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English majors at the University of Science - Vietnam National University - Ho Chi Minh City, using six modules (modules 07-12) in the book New Cutting Edge (Intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics namely everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, and finance-related issues. Students need to complete various tasks, including

	presentations, debates, role-plays, doing homework, tests and so on.
Examination forms	None
Study and examination requirements	Mid-term test: 50% Final test: 50%
Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, Intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, Intermediate: workbook.* Harlow: Pearson Education. 3. Materials prepared by the lecturer (2012). *Collins Skills for the TOEIC test: Speaking and Writing*. Harper Collins UK.

Environmental Studies - ENV00001

Module designation	Environmental Studies
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at Faculty of Environment, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	90 hours Contact hours: lectures 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This is a compulsory subject in the general knowledge block in the training program for students of all disciplines. The focus of this module is to provide basic knowledge of Environmental Science: concepts, classification of resources,

	environment, basic problems and core of the environment. Students are also introduced to measures to protect the environment, conserve resources towards the goal of sustainable development.
Content	Chapter 1: Overview of the Environment General concepts of environment Chapter 2: Basic composition of the environment (volumes) Chapter 3: Natural Resources 3.1 Definition 3.2 Classification 3.3 General issues of natural disaster Chapter 4: Human Impact on the environment 4.1 History of human impact on the environment 4.2 Human impact on environmental components Chapter 5: Environmental issues and sustainable development 5.1 Population and environment 5.2 Environmental pollution 5.3 Climate change 5.4 Sustainable development Chapter 6: Environmental management and Environmental Education
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Midterm test: 30% Final test: 70%
Reading list	Textbooks: 1. *Lecture on General Environment*. (in Vietnamese). compiled by a group of teachers. 2. Le Van Khoa (Editor), 2004. *Environmental Science* (in Vietnamese). Education Publishing House. References 1. Le Van Khoa, Doan Van Canh, Nguyen Quang Hung, Lam Minh Triet (2011). *Textbook of People and the*

	Environment. (in Vietnamese). Education Publishing House. 2. Goudie, A. (2006) *The Human Impact on Natural Environment*. 6th Edition. Oxford, Blackwell. 3. Le Thi Thanh Mai (2008), *Textbook of People and the Environment*. VNU-HCM Press.

General Biology I - BIO00001

Module designation	General Biology I
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Biology and Biotechnology, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	This course introduces the topic, gives questions, asks students to see the clip, discusses and reviews
Workload (incl. contact hours, self-study hours)	Total workload: 135 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 45 - Practical hours: 0 - Exercise hours: 0 - Extra-contact hours per week: 0 Private study including examination preparation, specified in hours: 90 hours
Credit points	3 Credits (4.5 ECTS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Know the fundamental concepts of biology and the scientific process; distinguish observation, hypothesis, test, and theory Explain the chemical composition of living matter, and chemical reaction related to biology Describe the structure and properties of water, structure and function of large biology molecules Distinguish the structure and function of cells, membranes and cellular organelles Describe metabolism at the cellular level, including enzyme activity, cellular respiration, and photosynthesis. Describe DNA structure and function, including replication and repair, transcription, and translation. Describe the mechanism of genetic: cell division, fertilization, mutation, and genetic variation. Explain Biotechnology Explain the evolution and biodiversity Describe the concepts and mechanisms of the origin of species
Content	Introduction to Biology: The fundamental concepts in biology The scientific methods to study the life Part 1. The chemical of life Part 2. The Cell Part 3. Genetics Part 4. Mechanisms of Evolution Conclusion
Examination forms	Multiple choice test

Study and examination requirements	Students create groups: listen, write, and give questions and answers. 6 Quizzes - Midterm examination - Final examination
Reading list	Textbooks 1. Bui Trang Viet (2012). *Cell Biology* (in Vietnamese). VNU-HCM Press. 2. Pham Thanh Ho (2011). *Biology: Cell, genetics and evolution* (in Vietnamese). VNU-HCM Press. Supplementary reading 1. Reece J. B., Urry, L. A., Cain M. L. 1., Wasserman S. A., Minorsky P. V., Jackson R.B., & Campbell N. A. (2014). *Campbell Biology* (1st edition), Pearson. 2. Lecture slides and related clips.

General Biology II - BIO00002

Module designation	General Biology II
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Biology and Biotechnology, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecturing, posing questions for discussions, presenting videos, giving homework
Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 135 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 45 - Practical hours: 0

	- Exercise hours: 0 - Extra-contact hours per week: 0 Private study including examination preparation, specified in hours: 90 hours
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: None Recommended prerequisites: General Biology I (BIO00001)
Module objectives/intended learning outcomes	PART 1. PLANT MORPHOLOGY AND FUNCTION Learners get to know about plant morphology, functions, and their ecological relationships. They explore nutrient absorption, material transport in vascular plants, and the vital role of soils in providing nutrients. Additionally, they examine plant life cycles, reproductive systems, and how plants respond to environmental cues for survival. PART 2. ANIMAL MORPHOLOGY AND FUNCTION Learners explore the correlation between animal anatomy and physiology. They delve into the structure and functions of key systems including digestion, circulation, respiration, immunity, endocrinology, reproduction, and the nervous system. Additionally, they study osmoregulation, excretion processes, and the mechanisms underlying animal sensation and movement. PART 3. ECOLOGY Learners explore ecological factors and their impact on organisms, studying adaptation strategies. They learn about ecosystem components, structural dynamics, functional processes, material and energy cycling, and equilibrium within ecosystems. Concepts of population dynamics, community interactions, and biosphere principles are also covered, along with the significance of biodiversity conservation for sustaining ecosystems locally in Vietnam and globally.
Content	Part 1. Plant Morphology and Function

	Part 2. Animal Form and Function Part 3. Ecology
Examination forms	Multiple choice test
Study and examination requirements	None
Reading list	1. Begon M., J.L. Harper, and C.R. Townsend (1986). *Ecology: Individuals, Populations and Communities*. Blackwell Scientific Publications. 2. Brooker, R.J, Widmaier E.P; Graham L.E and Stiling P.D. (214). *Biology*. McGraw-Hill. 3. Campbell N.A, J.B. Reece, L.A. Urry, M.I. Cain, S.A. Wasserman, P.V. Minorsky, and R. B. Jackson (214). *Biology* (1st Edition). Pearson, Benjamin Cummings. 4. Stiling P. D. (22). *Ecology: Theories and Applications* (Fourth Edition). Prentice-Hall, Inc. 5. Smith A.M; Coupland G; Dolan L; Harberd N; Jones J; Martin C; Sablowski R; and Amey A. (21). *Plant Biology*. Garland Science.

General Chemistry 1 - CHE00001

Module designation	General Chemistry 1
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 30 hours (in class) Exercise: 30 hours Private study: 90 hours (self-study)

Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This subject is the first Chemistry subject for students of Chemistry and Materials Science. The subject deals with the theoretical foundations of Chemistry related to the basic models of the atomic structure, the periodic changes in the properties of chemical elements, the fundamental forces of interaction in the matter, and the influence of chemical elements. their influence on the properties of matter in the solid, liquid, and gaseous states.
Content	1. Describe the structure of atoms and molecules. 2. Explain periodic changes in some properties of chemical elements. 3. Identify and distinguish basic types of chemical bonds. 4. Identify and explain the relationship between the fundamental forces of interaction in matter and the physical properties of matter.
Examination forms	Written exam Midterm test: 30% Final test: 70%
Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	Textbooks: 1. Nguyen Dinh Chi (2007). *General Chemistry* (in Vietnamese). Hanoi Education Publishing House. 2. Nguyen Dinh Soa (2000). *General Chemistry* (in Vietnamese). VNU-HCM Press. 3. Petrucci, R.H; Harwood, W.S; Herring, F.G (2002). *General Chemistry*. 8th Ed. USA: Prentice Hall. Others: 1. Le Thi So Nhu. Summary of General Chemistry lecture - internal documents (For internal circulation only).

General Chemistry 2 - CHE00002

Module designation	General Chemistry 2
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 30 hours (in class) Exercise: 30 hours Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Introduce students to the facts, concepts, and vocabulary of chemistry that will enable them to make informed decisions on issues that involve chemical concepts. Give opportunity to use the scientific method to formulate explanations for events in the natural world. Explore the role of measurement in the development of these explanations and the use of logic to test the validity of claims. Show relationships of chemistry to other disciplines, such as mathematics and the biological sciences. Demonstrate the application of chemistry concepts in the everyday world. Give opportunity to apply learned concepts in new situations of both practical and theoretical significance. Give opportunity to read selected scientific materials and apply appropriate language to describe and analyze scientific events. Develop students' scientific problem-solving skills through sequential, mathematically oriented assignments.
Content	The contents are chemical thermodynamics, chemical kinetics, general chemical equilibria, acid and base

	equilibria, pH, buffers, ionic compound solubility and equilibria electrochemistry.
Examination forms	Written exam Midterm test: 35% Quiz or Assignment: 15% Final test: 50%
Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	1. Ralph H. Petrucci, F. G. Herring, J. Madura, C. Bissonnette (2011), *General Chemistry, Principles and Modern Applications*. 10th Edition. Pearson Canada. 2. Nguyen Dinh Soa (2000). General Chemistry (in Vietnamese). VNU-HCM Press. 3. Petrucci, R.H; Harwood, W.S; Herring, F.G (2002). General Chemistry. 8th Ed. USA: Prentice Hall.

General Chemistry Laboratory 1 - CHE00081

Module designation	General Chemistry Laboratory 1
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Exercise: 60 hours Private study: 30 hours (self-study)
Credit points	2 Credits (4 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Chemistry 1 (CHE00001)

Module objectives/intended learning outcomes	1. Describe the chemical processes of experiments 2. Use the equipment in chemistry lab correctly 3. Calculate, write chemistry laboratory report correctly 4. Work in the lab safely, scientifically
Content	The objective of the general chemistry laboratory course is to become proficient in techniques used by practicing chemist, to use and handle safely and properly laboratory glassware, to carry out experiments safely and carefully in the laboratory, to obtain data accurately and to manipulate the data correctly. This course also complements and consolidates the theoretical knowledge acquired in the general chemistry lecture course. In as much as this course is only a supplement to general chemistry lecture courses, students must have had the general chemistry lecture.
Examination forms	Written exam Midterm test: 20% Experimental Reports: 60% Final test: 20%
Study and examination requirements	Minimum lab hours: 7/9 If score of the final exam ≤ 3, the final score will be (midterm test) *20% + (Average of Experimental Reports) *30% + (Final exam) *20%.
Reading list	1. General Chemistry Laboratory 1, Hoàng Ngọc Cường (2023). 2. General Chemistry Laboratory 1-Prelab-Report, Hoàng Ngọc Cường (2023).

General Chemistry Laboratory 2 - CHE00082

Module designation	General Chemistry Laboratory 2
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective

Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Exercise: 60 hours Private study: 30 hours (self-study)
Credit points	2 Credits (4 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Chemistry 2 (CHE00002)
Module objectives/intended learning outcomes	1. Describe the chemical processes of experiments 2. Use the equipment in chemistry lab correctly 3. Calculate, write chemistry laboratory report correctly 4. Work in the lab safely, scientifically
Content	General Chemistry Laboratory 2 is an introductory course to the general chemistry laboratory. The overall goal of this course is to introduce general chemistry focused on understanding the concepts within the labs and the scientific method. These concepts include but are not limited to: understanding basic operation in a chemistry laboratory, components of matter, stoichiometry, determination of concentration by titration, weight measuring, preparation and properties of inorganic and organic substances.
Examination forms	Written exam Midterm test: 20% + Experimental Reports: 60% + Final test: 20%
Study and examination requirements	Minimum lab hours: 7/9 If score of the final exam ≤ 3, the final score will be (midterm test) *20% + (Average of Experimental Reports) *30% + (Final exam) *20%.
Reading list	1. Hoang Ngoc Cuong, *General Chemistry Laboratory 2 Manual*. (in Vietnamese).

General Law - BAA00004

Module designation	General Law
Semester(s) in which the module is taught	1st semester

Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, seminars
Workload (incl. contact hours, self-study hours)	Total workload: 135 hrs Contact hours (lectures): 45 hrs Private study including examination preparation, specified in hours: 90 hrs
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: General Objective: Understand the basic legal concepts and terms related to the country's legal system and state apparatus; apply legal provisions to solve some simple case studies; help students form and develop some skills such as looking up legal documents, analyzing legal regulations, and working in groups, thereby improving their sense of survival, learning and working following the Constitution and regulations. The law, the right behavior orientation in life. Specific objectives/course output standards: Knowledge: Present basic legal concepts and terms related to the state apparatus and the Vietnamese legal system; Solve some exercise cases based on the provisions of a law book in the legal system of Vietnam. Skills: Analyzing legal regulations; Lookup legal documents; Working group. Attitude, diligence: Raise awareness of living, studying, and working following the Constitution and the law.
Content	The module provides knowledge about the structure of the State apparatus as well as the functions, authority, and legal status of agencies in the State apparatus of the Socialist

	Republic of Vietnam in terms of economic management, Legal nature, and structure of the system of legal documents. From an overview of the system of legal branches in our State's legal system, a course is devoted to studying the basic contents of administrative law, civil law, and criminal law as branches of law. the main law (original branches of law) of the legal system, so that learners can easily access themselves to other branches of law arising from these major branches of law.
Examination forms	Written exam, Multiple choices, Oral presentation
Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	1. *General Law textbook*, Ho Chi Minh City University of Law. 2. *Textbook of Theory of State and Law*, Hanoi University of Law.

General Physics 1 (Mechanics and Thermodynamics) - PHY00001

Module designation	General Physics 1 (Mechanics and Thermodynamics)
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Prof. Chau Van Tao (Faculty of Physics and Engineering Physics, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, discussion.
Workload	135 Hours Contact hours: Lectures: 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ETCS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	This course covers the principles of kinematics, dynamics, statics, work, energy, linear momentum, gravitation, and thermodynamics. Students who complete this module could achieve the following: Knowledge: Be able to understand and apply laws of mechanics to explain physical phenomena and solve problems; Be able to understand and apply mechanisms of heat transfer, equations of state, the first and the second law of thermodynamics. Skills: Be able to work at an individual level and group work. Competence: Ability to apply mechanics and thermodynamics knowledge to analyze physical situations. Attitude: Honest
Content	This module includes the following topics: 1. Physics and measurement 2. Kinematics of particles 3. Force and Newton's laws 4. Conservation laws in classical mechanics 5. Kinetics of rigid bodies 6. The ideal gas 7. The first law of thermodynamics 8. The second law of thermodynamics
Examination forms	Class discussion. Quizzes and projects. Mid-term and Final exam: Written exam (closed book).
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final score is greater or equal to 5.0/10.0
Reading list	Main books: 1. Nguyen Nhat Khanh (2005). *Mechanics and thermodynamics lectures*. VNU-HCM Press. References: 1. Nguyen Thanh Van. (2013). *General Physics 1*. VNU-HCM Press.

	2. Raymond A. Serway, John W. Jewett, Sr, (2014). *Physics for Scientists and Engineers with Modern Physics*. Brooks/Cole Publishing Company, USA. 3. Alan Giambattista, Betty McCarthy Richardson, Robert C. Richardson, (2010). *Physics*. McGrawHill Companies, Inc, USA.

General Physics 2 (Electromagnetism - Optics) - PHY00002

Module designation	General Physics 2 (Electromagnetism - Optics)
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Assoc. Prof. Huynh Truc Phuong (Faculty of Physics and Engineering Physics, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lesson, discussion
Workload	135 Hours Contact hours: Lectures: 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Physics 1 (PHY00001)
Module objectives/intended learning outcomes	This module provides basic knowledge of electric and magnetic fields and thereby an understanding of the laws and phenomena of light optics. Students who complete this module could achieve the following: Knowledge: Be able to understand and apply knowledge of electromagnetism and optics in science and life. Skills: Be able to work at individual level and teamwork. Competence: Ability to apply electromagnetism and optics knowledge to analyze physical situations. Attitude: Honesty and diligence

Content	This module includes the following topics: 1. Electric charge and electric field 2. Conductors in an electric field 3. Electric current and magnetic field 4. Electromagnetic induction and applications 5. The background of light optics 6. Interference of light 7. Diffraction of light 8. Polarization of light
Examination forms	Oral presentation Mid-term and Final exam: written exam (closed book)
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final score is greater or equal to 5.0/10.0
Reading list	Main books: 1. Nguyen Thanh Van. (2015). *General Physics*. (in Vietnamese). VNU-HCM Press. References: 1. Le Vu Tuan Hung (2015). *Optics*. (in Vietnamese). VNU-HCM Press. 2. Raymond A. Serway, John W. Jewett, Sr (2014). *Physics for Scientists and Engineers with Modern Physics*. 9th Edition. BROOK/COLE, USA. 3. Alan Giambattista, Betty McCarthy Richardson, Robert C. Richardson (2010). *Physics*. 2nd Edition. McGrawHill, USA.

History of Vietnamese Communist Party - BAA00104

Module designation	History of Vietnamese Communist Party
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory

Teaching methods	Lecture, lesson, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	In terms of content: providing systematic and basic knowledge about the birth of the Communist Party of Vietnam (1920-1930), the Party's leadership over the Vietnamese revolution during the period of political struggle. government authority (1930-1945), in two resistance wars against French colonialism and American imperialism (1945-1975), in the cause of national construction and defense during the country's transition to socialism. association, conducting renovation work (1975-2018). Ideologically: Through historical events and experiences on the leadership of the Party to build a sense of respect for objective truths, raise pride and confidence in the Party's leadership cause. About skills: Equipping with scientific thinking methods on history, skills in choosing research materials, studying subjects and the ability to apply historical awareness to practical work, criticizing misconceptions on the history of the Party.
Content	The course provides systematic and fundamental knowledge about the birth of the Communist Party of Vietnam (19201930), the Party's leadership for the Vietnamese revolution during the period of struggle for power (1930-1945), during the two resistance wars against the French colonialists and the American imperialists (1945-1975), in the cause of national construction and defense during the period of the whole country's transition to socialism, conducted doi moi (1975-2018). Through historical events and experiences in the leadership of the Party to build a

	sense of respect for objective truth, heighten pride and confidence in the Party's leadership cause. Equip with scientific thinking methods on history, skills in choosing research materials, studying subjects, and the ability to apply historical awareness to practical work, and criticize misconceptions about the history of the Party.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations for group presentations Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam. Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have test scores, lively discussions, constructive and serious statements in class.
Reading list	*Curriculum of the History of the Communist Party of Vietnam*, Issued by the Ministry of Education and Training.

Ho Chi Minh's Ideology - BAA00003

Module designation	Ho Chi Minh's Ideology
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM

Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, lesson, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures:30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	About knowledge: Equip students with basic knowledge about the concept, origin, the process of formation and development of Ho Chi Minh thought; the basic contents of Ho Chi Minh's thought; the application of the Communist Party of Vietnam in the national-democratic revolution and the socialist revolution, in the current national renewal process. About skills: Helping students to think, analyze, evaluate, and creatively apply Ho Chi Minh's Thoughts to solve problems in real life, study, and work. About attitudes: Helping students improve their political bravery, patriotism, loyalty to the goal, the ideal of national independence associated with socialism; aware of the role and value of Ho Chi Minh's thought for the Vietnamese Party and nation; realize their responsibility in studying and training to contribute to the construction and defence of the country.
Content	Description of course content: the subject equips students with basic knowledge about objects, research methods, and learning meanings of Ho Chi Minh's ideology; on the basis, of the process of formation and development of Ho Chi Minh thought; on national independence and socialism; on the Communist Party and the State of Vietnam; on great national and international solidarity; about culture, ethics, people.

Examination forms	Class discussion. Group presentations and reports. Mid-term exam: Multiple choice (closed-book) or essay (opened-book). Final exam: Essay (opened book).
Study and examination requirements	Regulations for group presentations Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam. Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance, and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have test scores, lively discussions, constructive and serious statements in class.
Reading list	1. *Textbook of Ho Chi Minh's Thoughts*, National Political Publishing House of Vietnam. 2. *Study Guide for Ho Chi Minh's Thoughts*. VNU-HCM Press.

Human and Environment - ENV00003

Module designation	Human and Environment
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Le Cong Man (Faculty of Biology and Biotechnology, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, disscusion

Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 30 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 30 - Practical hours: 0 - Exercise hours: 0 - Extra-contact hours per week: Private study including examination preparation, specified in hours: hours
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: General Biology Recommended prerequisites: 0
Module objectives/intended learning outcomes	Providing students with knowledge about humans' relationship with nature, including inanimate factors and other living things around humans Knowledge: knowing the position, tasks, and roles of humans in the ecosystem. Skills: recognizing results from human activities and impacts on natural and social life. Attitude, Diligence Able to practice soft skills (teamwork, presentations)
Content	Part 1: The formation of human community and society Chapter 1: Introduction Chapter 2: Ecological factors Chapter 3: Population - biome - ecosystem Chapter 4: People and the environment Chapter 5: History of human impact on the environment Part 2: Needs and products in human society Chapter 6: Essential human needs Chapter 7: Resources

	Chapter 8: Pollution Chapter 9: Environmental protection and sustainable development Chapter 10: Cultural and legal issues. Part 3: Observe and identify (from an ecological point of view) ongoing phenomena affecting the lives of humans and other living things.
Examination forms	Mini projects
Study and examination requirements	Study Requirements: 0 Examination Requirements: 1. Discuss 2. Teamwork 3. Seminar 4. Learning attitude 5. Final exam
Reading list	1. Pham Xuan Hau (1997). *Human and Environment*. (in Vietnamese). Vietnam Education Publishing House. 2. Le Van Khoa (2002). *Environmental Science*. (in Vietnamese). Vietnam Education Publishing House. 3. Nguyen Huu Nhan, Hoang Quy Tinh (2011). *Human Ecology Textbook*. (in Vietnamese). Vietnam Education Publishing House.

Introduction to Informatics - CSC00003

Module designation	Introduction to Informatics
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at Centre of Informatics, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, problem, lab works
Workload (incl. contact hours, self-study hours)	Total workload: 165 hrs Contact hours:

	- Lecture hours: 15 - Practical hours: 60 - Exercise hours: 0 Private study: 90 hours (self-study)
Credit points	3 Credits (5.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: 1. Explain common concepts and terms related to the field of computer science. 2. Perform basic operations related to the Windows operating system and common software. 3. Create documents with professional format, structure, and presentation using software. 4. Analyze and organize data in spreadsheet format, use calculation, search, and statistical functions to create formulas for data processing and presentation. 5. Create eye-catching presentations. 6. Express concepts related to the Internet, information security on the network. 7. Be aware of searching for information and legal regulations regarding copying, sharing, and posting information on the Internet. 8. Design a basic electronic information page.
Content	• Basic understanding of IT • Basic knowledge of computers and computer networks. • Access control, ensuring data security, Malware. • Basic legal issues in using IT. • Basic computer usage • Windows operating system • Windows Explorer • Control Panel • Data compression and extraction • Typing in Vietnamese • Basic Microsoft Word • Document composition • Text formatting • Creating tables • Handling graphics in documents • Page layout and printing

	• Basic Microsoft PowerPoint • Basic presentation templates • Creating a presentation • Setting up effects for the presentation • Basic Microsoft Excel • Data formatting in Excel • References in Excel • Basic Excel functions • Printing and creating charts • Internet usage • Basic knowledge of the Internet • Information searching • Information security • Web image processing • Resizing image frames • Image cropping • Image rotation and flipping • Adjusting the brightness of an image • Adjusting the contrast of an image • Designing a basic electronic information page using HTML & CSS3.
Examination forms	Multiple choice exam, practical exam
Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	1. *Curriculum for Basic IT Applications*, Advanced IT Applications. 2. *Microsoft Office MOS materials*, IIG Vietnam, Fahasha. 3. *IC3 Spark materials*, IIG Vietnam, Fahasha.

Labwork on General Biology I - BIO00081

Module designation	Labwork on General Biology I
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Luong Thi My Ngan (Faculty of Biology and Biotechnology, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective

Teaching methods	Lecture Introducing laboratory rules Divide students into groups of three or two Guide experimental procedure Monitor and examine students doing experiments
Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 30 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 0 - Practical hours: 30 - Exercise hours: 0 - Extra-contact hours per week: Private study including examination preparation, specified in hours:
Credit points	1 Credits (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Biology I (BIO00001)
Module objectives/intended learning outcomes	Prepare temporary slides to observe cell structures and living activities. Use a microscope proficiently. Identify organic compounds (carbohydrates, proteins, lipids, enzymes like amylase). Demonstrate and measure plant respiration. Classify plants (lichens, algae, bryophytes, pteridophytes, gymnosperms, monocots, dicots). Classify animals into major orders. Use basic lab tools (test tubes, beakers, pipettes, etc.) effectively. Collaborate and assist team members to complete tasks on time. Follow lab safety rules and ensure personal and team safety.

Content	Introduction of Syllabus How to use a light microscope Cell structure; Osmosis; Chemical Composition of Cells; Enzyme Activity; Respiration; Mitosis; Microbial Diversity. Plant Diversity; Animal Diversity
Examination forms	Lab Activity and Lab Report Lab exam
Study and examination requirements	Study Requirements: Listen Do experiments Making lab report Put tools back in right places Clean up. Examination Requirements: Experimental process & Final exam
Reading list	Main textbook: 1. Lab manual: Bui Trang Viet & Le Thi My Phuoc (Editors) (2016), *Lab work on General Biology I.* (Internal circulation) 2. Class lecture (ppt file) for *Lab work on General Biology 1.* (in Vietnamese). Supplementary: 1. Dickey Jean (2003). *Laboratory Investigations for Biology*. Pearson.

Labwork on General Biology II - BIO00082

Module designation	Labwork on General Biology II
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	MSc. Phan Ngo Hoang (Faculty of Biology and Biotechnology, University of Science, VNU-HCM)

Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	preaching, guiding, discussion
Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 30 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 0 - Practical hours: 30 - Exercise hours: 0 - Extra-contact hours per week: Private study including examination preparation, specified in hours:
Credit points	1 Credit (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Biology II (BIO00002)
Module objectives/intended learning outcomes	1. Distinguish different plant tissues and organs. 2. Understand and classify plant pigments. 3. Demonstrate environmental impacts on photosynthesis. 4. Explore respiration and fermentation in plants. 5. Identify different animal tissues and prove bioelectricity. 6. Explain hormonal effects on blood flow. 7. Understand climatic impacts on ecosystems. 8. Describe biological reactions and processes. 9. Interpret respiration and photosynthesis. 10. Explain cellular signal transduction. 11. Distinguish plant and animal structures and evolution. 12. Conduct and analyze biological experiments and biodiversity research.
Content	1. Practice theory 2. Tissue and plant organs 3. Plant pigments

	4. Photosynthesis 5. Respiration and fermentation in plants 6. Water transport of tissues 7. Animal tissues 8. Mechanical and bioelectric activities 9. Circulation of blood in the vein 10. Climate factor
Examination forms	Abstract paper: objectives, principles and methods of practice Practice report Practice and question
Study and examination requirements	Listen, take note, practising, discussing Students must not be absent in practice theory class (the first week) Students wear blouses when entering the Lab Submit practice reports at the end of the practice session Students who are absent for more than 2 practice sessions will be banned from the final exam. Examination Requirements: Homework Lab work Final examination
Reading list	Textbooks: 1. Bui Trang Viet, Phan Ngo Hoang (2009). *Practical Textbook for General Biology 2* (in Vietnamese). VNU-HCM Press. Supplementary reading: 1. Lecture notes. 2. Pham Thanh Ho (1998). *General Biology.* (in Vietnamese). Vietnam Education Publishing House.

General Physics Lab - PHY00081

Module designation	General Physics Lab
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	MSc. Huynh Thanh Nhan (Faculty of Physics and Engineering Physics, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lab work, discussion, practice
Workload (incl. contact hours, self-study hours)	120 Hours Contact hours: Lab work: 60 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (4 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course is a practical subject in the laboratory. This course helps students understand how to measure some physical quantities, experimental errors, analyze and evaluate measurement results. Students who complete this module could achieve the following: Knowledge: Be able to describe the process, how to measure fundamental physical quantities in the laboratory. Be able to use instruments and equipment to measure experimental data of physical quantities correctly. Be able to determine (calculate) physical quantities from measured experimental data. Be able to determine the error of experimental measurement of physical quantities. Skills: Be able to work in individual, group work, self-study, and problem solving. Competence: Be able to analyze, process and write experimental data reports. Attitude: be honest, responsible, respect for colleagues.
Content	In this module, Students practice 10 of the following 13 experiments:

	Practice 1: Density of liquid and solids. The private mass of the metals. Practice 2: Viscosity. Viscosity is dependence of different temperatures. Practice 3: Reversible pendulum. The Mathematical pendulum. Practice 4: Heat of function for ice. Determination of heat. Practice 5: Mechanical equivalent of heat. The heat capacity of metals. Practice 6: Wheatstone Bridge. Resistor is dependent of different temperatures. Practice 7: Voltmeter and Ammeter DC. Voltmeter and Ammeter AC. Practice 8: AC circuit. RLC circuit. Practice 9: Diode characteristics. Practice 10: Transistor characteristics. Practice 11: Microscope. To measure the diameter of other small objects. Practice 12: Refraction by a prism. Dispersion and resolving the power of prisms. Practice 13: Polarization of light Rotatory power.
Examination forms	Practice reports, practice exam
Study and examination requirements	Minimum attendance at Lab is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final score is greater or equal to 5.0/10.0
Reading list	1. Dang Van Liet, Do Dinh Luyen, Nguyen Van Nghia, Tran Thi Kim Phuong (2008). *General Physics Experiments*. (in Vietnamese). VNU-HCM Press.

Marxist-Leninist Philosophy - BAA00101

Module designation	Marxist-Leninist Philosophy

Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The course equips students with the basic contents of the worldview and the Marxist-Leninist philosophical methodology. Helping students apply knowledge about the worldview, Marxist-Leninist philosophy, and philosophy creatively in cognitive and practical activities, to solve problems that the social life of a country or of the time being set.
Content	Marxist-Leninist philosophy is a course in the Marxist-Leninist knowledge block and Ho Chi Minh Thought. This module equips students with basic, general, and systematic knowledge of the problems of Marxist-Leninist philosophy. From there, learners have a basis and reasonable research and learning methods and apply them to the process of evaluating life phenomena.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book)
Study and examination requirements	Regulations for group presentations

	Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam. Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have test scores, lively discussions, constructive and serious statements in class.
Reading list	1. *Textbook of basic principles of Marxism-Leninism*, National Political Publishing House of Vietnam. 2. *Textbook of Marxist-Leninist Philosophy*, National Political Publishing House of Vietnam.

Marxist-Leninist Political Economy - BAA00102

Module designation	Marxist-Leninist Political Economy
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Firstly, equip students with basic and core knowledge of the Marxist-Leninist political economy in the context of economic development of the country and the world today. Ensure the basic, systematic, scientific, updating new knowledge, associate with practice, creativity, skills, thinking, learner quality, connectivity to overcome duplication, enhance integration and reduce the load, reduce content that is no longer relevant or scholastic content for students at non-theoretical colleges and universities. Second, on that basis, forming thinking and analytical skills, assessing and identifying the nature of economic benefit relations in the country's socio-economic development, contributing to helping students build appropriate social responsibility in the job position and life after graduation. Third, contribute to building the stance and ideology of Marxism-Leninism towards students.
Content	The program consists of six chapters: in which chapter one discusses the objects, research methods, and functions of the Marxist-Leninist Political Economy. Chapters 2 to 6 present the core content of the Marxist-Leninist political economy according to the subject's objectives. Specifically, issues such as goods, markets and the role of actors in the market economy; Producing surplus value in a market economy; Competition and monopoly in the market economy; Socialist-oriented market economy and economic interest relations in Vietnam; Industrialization, modernization, and international economic integration in Vietnam.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations for group presentations: Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam.

	Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance, and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have all test scores, lively discussions, constructive and serious statements in class.
Reading list	Marxist-Leninist political economy textbook for undergraduates who are not majoring in political economy.

Psychology - BAA00006

Module designation	Psychology
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Tran Huong Thao
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total workload: 90 hours. Contact hours (lecture): 30 hours. Private study including examination preparation, specified in hours: 60 hours.
Credit points	2 credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Knowledge:

	• Understand the system of basic concepts of psychological science and research methods in psychology. • Understand the origin, formation and development of psychology and consciousness. • Understand the nature of human psychological processes: perception; emotion - affection; act. • Identify human psychological states. • Understand the psychological attributes that make up the personality structure. Understand the factors affecting the formation and development of personality. Skills: • Developing the capacity to study documents: Analyze, synthesize, compare, and generalize. • Form and develop the ability to identify psychological phenomena and apply learned knowledge to solve practical problems. • Consulting and consulting skills. Attitude: • Cultivate a passion for learning and studying subjects. • Forming a sense of initiative and positivity in self-study. • Form the right motivation in learning. • Raise a sense of responsibility for group activities. • Other goals: • Forming personality qualities in accordance with the requirements of the integration period. • Forming communication and behavioral skills in the community. • Forming a modern and scientific way of living and working. • Forming and developing the ability to think creatively, independently, and critically. • Reasoning skills, public speaking skills. • Form and develop teamwork skills.
Content	The course introduces learners to acquire basic knowledge about the nature and characteristics of psychological phenomena and basic psychological laws of humans (perception, emotion, will, etc.) actions and personalities...). On that basis, it helps learners to apply knowledge in

	practice to identify and distinguish basic psychological phenomena in humans. This module includes the following parts: Part 1: The Natural Basis and Social Foundation of Human Psychology. Part 2: Perception and Learning. Part 3: Emotions - Affections. Part 4: Personality and the Formation of Personality. Part 5: Deviations in Individual Psychological Behavior and Correcting Such Deviant Behavior.
Examination forms	Class discussion; Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations on time, attendance, and discipline in the course: attend class on time and at least 70% of the sessions (only to be absent for a maximum of 30%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have all test scores, lively discussions, constructive and serious statements in class.
Reading list	Main textbook: 1. Nguyen Quang Uan (2015). *General Psychology: A Textbook*. (in Vietnamese). References: 1. Plotnik, R, Kouyoumdjian, H (2011). *Introduction to Psychology*. 2. Berstein, D. A., Penner, L. A., Clarke-Stewart, A., and Roy, E (2008). *Psychology*.

Scientific Socialism - BAA00103

Module designation	Scientific Socialism
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at School of Political and Administration

	Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	lecture, lesson, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The subject equips students with the basic contents of scientific socialism (one of the three components constituting Marxism-Leninism). Helping students apply basic knowledge of scientific socialism creatively in cognitive and practical activities, solving problems that the social life of a country, of the times being set.
Content	The subject equips students with the basic contents of scientific socialism (one of the three components constituting Marxism-Leninism). Helping students apply basic knowledge of scientific socialism creatively in cognitive and practical activities, solving problems that the social life of a country, of the times being set.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations for group presentations Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam.

	Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures)
Reading list	1. *Textbook of Scientific Socialism*, National Political Publishing House of Vietnam. 2. *The Basic Principles of Marxism-Leninism*, National Political Publishing House of Vietnam.

Teamwork and Learning Skills - BAA00008

Module designation	Teamwork and Learning Skills
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Tran Huong Thao
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 60 hours Contact hours: Lectures 30 hours (in-class). Private study: 30 hours (self-study).
Credit points	2 credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: Specific objectives/subject output standards: Equip students with knowledge and skills about teamwork and

	study skills. Bringing career orientations to group work practice topics. Skills: teamwork, presentation, communication Attitude, diligence: serious, diligent, positive
Content	Understand the concept of groups, how to form groups, classify groups, functions, and tasks of group members. Knowing the process of teamwork, the necessary skills when working in groups. Able to plan and execute projects. Understand and apply learning skills.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Team Leadership Techniques*. Tre Publishing House. 2. *Strategic Learning*. Labour & Social Publishing House.

Analysis 1A - MTH00010

Module designation	Analysis 1A
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Assoc. Prof. Dr. Ly Kim Ha
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: lectures: 30 hours (in class); 30 exercise hours Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	The objective of the module is to equip students with the basic knowledge of the foundation of calculus as the foundation for specialized modules.
Content	The course covers the basics of real numbers, sequences and series of real numbers, one variable continuous function.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Mathematica by example*, Academic Press, New York 2. *Calculus*, Harcourt Brace College Publishers, New York 3. *Textbook of Calculus I*. Statistical Publishing House, Ho Chi Minh City. 4. *Analysis 1*, Tenence Tao, 2022, Sringer.

Analysis 2A - MTH00012

Module designation	Analysis 2A
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Le Anh Ha
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 30 hours *(in class).* Exercises: 30 hours *(in class)* Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites • Analysis 1A (MTH00010)

Module objectives/intended learning outcomes	General objective: To introduce the basic concepts and techniques of metric topology for multivariable functions and vectors. Specific objectives: Knowledge: know the basic concepts of metrics, closed, open, compact sets, completeness, series in normed space. Skills: do exercises that demonstrate the above concepts and their applications
Content	This module helps students understand basic topological concepts, understand convergence in multi-dimensional spaces and function spaces. The course introduces metric spaces, normative spaces and Rn, properties of continuous functions on metric spaces. Completeness of spaces and series on complete normed space. This knowledge forms the basis for all specializations such as analysis, statistics and probability, computer science, optimization, and applied mathematics.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 20%, Midterm Exam: 30%, Final Exam: 50%
Reading list	1. Dang, D. T., Dinh, N. T., Pham, H. Q. (2005). *Calculus A2* (in Vietnamese). VNU-HCM Press. 2. Tenence Tao. (2022). *Analysis 2*. Springer. 3. Nancy, G. P., Scott, F. (1985). *Calculus*. Harcourt Brace College Publishers, New York.

Analysis 3A - MTH00014

Module designation	Analysis 3A
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Huynh Quang Vu
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	Lectures, group work, small group solving exercises

Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: lectures 45 hours + 30 exercises hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Analysis 1A (MTH00010) • Calculus 1A (MTH00011) • Analysis 2A (MTH00012)
Module objectives/intended learning outcomes	General objective: Study the integral of functions of many variables and the relationships between the differential and integral of functions of many variables. Specific objectives: Knowledge: The minimum outcome standard is at the level in J. Stewart's Calculus textbook for science and engineering students. The average level is more advanced, more suitable for Mathematics majors, with higher requirements for accuracy and theoretical content. For fairly good students and honor students the course aims for qualifications in the respective sections of classic analysis textbooks such as those of W. Rudin, S. Lang. Skills: Introduction to computer tools. Exercises include both reasoning and calculation. Attitude, diligence: Seeing the need to develop generalization and precision, forming an ability to solve new application problems. Having a serious, proactive and self-disciplined learning attitude.
Content	This is a course on Multiple Integration and Vector Calculus. This subject follows the subjects of Calculus 1 and Calculus 2, which is considered a basic knowledge for university level in Science and Technology. A useful course for more advanced investigations of Lebesgue integrals (Measurement and Probability), mathematical models using Integral (in Mechanics, Probability-Statistics, Mathematical Equations, Calculus, ...), and mathematical developments (in Analysis, Geometry, ...)

Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 20%, Midterm Exam: 30%, Final Exam: 50%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Calculus III Textbook*, Huynh Quang Vu, VNU-HCM Press, 2017. Class notes available at https://sites.google.com/view/hqvu/teaching 2. *Calculus: Early Transcendentals*. James Stewart, 8th ed., Brooks/Cole, 2016. 3. *Advanced Calculus*. Gerald B. Folland, Pearson, 2001.

Ordinary Differential Equations - MTH00015

Module designation	Ordinary Differential Equations
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Dr. Le Anh Ha
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	135 hours Contact hours: lectures 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Analysis 1A (MTH00010) • Calculus 1A (MTH00011) • Analysis 2A (MTH00012) • Calculus 2A (MTH00013)
Module objectives/intended learning outcomes	General objective: To introduce basic differential equations and know how to solve basic ordinary differential equations. Specific objectives/course outcome standards:

	Knowledge: Master the course content. Skills: Ability to calculate with large numbers of calculations Attitude, diligence: Diligent and serious
Content	This module equips math majors with a minimum of background knowledge before entering narrower specializations. The content of this course includes: Differential equations of first order. Existence and uniqueness of solutions of Cauchy problem. Linear differential equations of second and higher order. Introduction to the system of differential equations of first order.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Midterm Exam: 20%, Final Exam: 80%
Reading list	1. Nguyen Thanh Long (2012). *Lecture Notes on Calculus A4* (in Vietnamese). 2. Nguyen, T. L., Nguyen, C. T., Le, T. P. N., and Nguyen, A. T. *Differential Calculus of Multivariable Functions and Differential Equations* (in Vietnamese). 3. Ravi, P. A., Donal, O'R. (2008). *An introduction to ordinary differential equations*. Springer.

Calculus 1A - MTH00011

Module designation	Calculus 1A
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Assoc. Prof. Dr. Ly Kim Ha
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours:

	Lectures: 30 hours (in class); 30 exercise hours Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The objective of the module is to equip students with the basic knowledge of calculus as the foundation for specialized modules.
Content	The course covers the basics of classic differentiation, and Riemann integrations.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Mathematica by example*, Academic Press, New York 2. *Calculus*, Harcourt Brace College Publishers, New York 3. *Textbook of Calculus I*. Statistical Publishing House, Ho Chi Minh City 4. *Analysis 1*, Tenrence Tao, 2022, Springer.

Calculus 2A - MTH00013

Module designation	Calculus 2A
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Le Anh Ha
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: lectures: 30 hours (in class); 30 exercise hours Private study: 90 hours (self-study)

Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Calculus 1A (MTH00011)
Module objectives/intended learning outcomes	General Objective: To introduce the basic concepts and techniques of differentiable calculus of multivariable vector functions. Specific objectives: Knowledge: The theory of a series of real numbers. Properties of differentiable functions on Rn. Skills: be able to do calculation exercises on the above concepts and their applications
Content	This module helps students understand the basic knowledge of differential calculus of multivariable functions, understand bound and unconstrained extremal problems in multidimensional spaces, and understand the concept of series of real numbers. This knowledge forms the basis for all specializations such as analysis, statistics and probability, computer science, optimization, applied mathematics.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 20%, Midterm Exam: 30%, Final Exam: 50%
Reading list	1. Dang, D. T., Dinh, N. T., Pham, H. Q. (2005). Calculus A2 (in Vietnamese). VNUHCM Press. 2. Nancy, G. P., Scott, F. (1985). Calculus. Harcourt Brace College Publishers, New York. 3. Sudhir, R. G., Balmohan, V. L. (2010). A Course in Multivariable Calculus and Analysis. Springer.

Computational Software Laboratory - MTH00087

Module designation	Computational Software Laboratory
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Ong Thanh Hai
Language	Vietnamese
Relation to curriculum	Compulsory

Teaching methods	Lectures, computer practices
Workload (incl. contact hours, self-study hours)	120 hours Contact hours: 60 lab hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (4 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Students can use computational programming languages to program calculations in mechanics, analysis, numerical analysis, algebra, statistics…
Content	The course covers basic computational programming knowledge using Matlab... Applying this knowledge to write programs to solve problems in mechanics, analysis, numerical analysis, algebra, ...
Examination forms	Tests - process score: 30% Midterm exam: 30% Final exam: 40%
Study and examination requirements	Midterm and final exam: practical exercises in the laboratory. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Brian, D. H., Daniel, T. V. (2007). *Essential MATLAB ® for Engineers and Scientists*, 3rd edition, Elsevier. 2. J. H. Mathews, K. D. Frink, (1999). *Numerical methods using Matlab*. 3rd Edition. Prentice Hall. 3. Ronald, L. L., Jonathan, M. R. (2017). *Multivariable Calculus with MATLAB: With Applications to Geometry and Physics*. Springer.

Linear Algebra - MTH00030

Module designation	Linear Algebra
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Dr. Le Van Luyen

	Dr. Bui Anh Tuan
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	135 hours Contact hours: lectures 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: To be familiar with advanced mathematics. Specific objectives/course learning outcomes: Knowledge: Mastering the knowledge of matrices on number fields and applying them to solving systems of linear equations; determinants and their applications; vector spaces and linear transformations. Skills: Calculating matrices; solving system of linear equations; calculating the coordinates of the vector according to a basis in a finite dimensional vector space; changing when changing the base; representing a linear operator by a matrix; finding the image and the kernel of linear operators; using MAPLE software. Attitude: Attend all classroom sessions; discussions outside of class time.
Content	This course is taught in the first semester, initially introducing students to advanced mathematics. Besides equipping new knowledge that is necessary for freshman, this course also provides the foundation knowledge to help students carry out specialized courses.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Midterm: 40% Final: 60%

Reading list	1. Bui, X.H., Tran, N.H., Trinh, T.D, Le, V. L. (2009). *Linear Algebra and Its Applications*, Vol 1 (in Vietnamese). VNUHCM Press. 2. Ngo, V. T. (2001). *Textbook of Linear Algebra*. (in Vietnamese). Vietnam National University Press, Hanoi. 3. David, C. L. (2011). *Linear Algebra and Its Applications*. (in Vietnamese). Pearson Education.

Linear Algebra Practice - MTH00083

Module designation	Linear Algebra Practice
Semester(s) in which the module is taught	1st semester
Person responsible for the module	MSc. Pham The Nhan
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	60 hours Contact hours: exercises 30 hours (in class) Private study: 30 hours (self-study)
Credit points	1 credit (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030)
Module objectives/intended learning outcomes	General objectives: To help students understand more linear algebra via doing exercise. Specific objectives/course learning outcomes: Knowledge: Mastering the knowledge of matrices on numerical fields and applying them to solving systems of linear equations; determinants and their applications; vector spaces and linear transformations. Skills: Calculating matrices; solving system of linear equations; calculating the coordinates of the vector according to a basis in a finite dimensional vector space; changing when changing the base; representing a linear

	operator by a matrix; finding the image and the kernel of linear operators; using MAPLE software. Attitude: Attend all classroom sessions; discussions outside of class time.
Content	This course is taught in the first year, initially introducing students to advanced mathematics. Besides equipping new knowledge that is necessary for freshman, this course also provides the foundation knowledge to help students carry out specialized courses.
Examination forms	Tests - process score: 30% Midterm exam: 30% Final exam: 40%
Study and examination requirements	Midterm and final exam: written exams.
Reading list	1. Bui, X.H., Tran, N.H., Trinh, T.D, Le, V. L. (2009). *Linear Algebra and Its Applications*, Vol 1 (in Vietnamese). VNUHCM Press. 2. Ngo, V. T. (2001). *Textbook of Linear Algebra*. (in Vietnamese). Vietnam National University Press, Hanoi. 3. Pham, H. D. (2009). *Computer, Program and Teach Mathematics on Maple.* (in Vietnamese). Science and Technics Publishing House.

Abstract Algebra - MTH00031

Module designation	Abstract Algebra
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Trinh Thanh Deo
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	135 hours Contact hours: lectures 45 hours (in class) Private study: 90 hours (self-study)

Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030) • Linear Algebra Practice (MTH00083)
Module objectives/intended learning outcomes	The goal of the course is to equip students with basic abstract algebraic structures, to help them become familiar with symbols and formal calculations.
Content	The course introduces basic algebraic structures such as groups, rings, integer domains, fields, and polynomial rings on fields. Some special concepts introduced are permutation group, alternating group, cyclic group, polynomial ring on number fields, especially on rational number fields.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nguyen, V. D., Tran N. H. (2005). *Abstract Algebra* (in Vietnamese). VNUHCM Press. 2. Hoang, X. S. (1997). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House. 3. Nguyen, H. V. H. (1998). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House. 4. My. V. Q. (1998). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House. 5. Bui, H. H., Nguyen H. H., Phan D. T. (1985), *Algebra and Arithmetic Exercises*, Vol 1 and 2. (in Vietnamese). Vietnam Education Publishing House.

Abstract Algebra Practice - MTH00084

Module designation	Abstract Algebra Practice
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Nguyen Kim Ngoc
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	lectures, computer practices

Workload (incl. contact hours, self-study hours)	90 hours Contact hours: practice 30 hours (in lab) Private study: 60 hours (self-study)
Credit points	1 Credit (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030) • Linear Algebra Practice (MTH00083) • Abstract Algebra (MTH00031)
Module objectives/intended learning outcomes	General Objective: To equip students with basic computational skills in abstract algebraic structures, formal calculation, giving them a better understanding of these algebraic structures. Use softwares (e.g. GAP) to calculate specific examples and exercises. Specific objectives/course learning outcomes: Knowledge: Master the theory of basic algebraic structures such as groups, rings, fields, and polynomial rings. Get hands-on with software (e.g. GAP) to better understand the structure learned. Skills: Improve reasoning skill and formal calculation. Attitude, diligence: Serious, progressive.
Content	The goal is to practice basic computational skills on algebraic structures such as groups, rings, integral domains, and fields, and use software (e.g. GAP) to better understand the structure of theories learned.
Examination forms	Tests - process score: 30% Midterm exam: 30% Final exam: 40%
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nguyen, V. D., Tran N. H. (2005). *Abstract Algebra* (in Vietnamese). VNUHCM Press. 2. Hoang, X. S. (1997). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House.

	3. Alexander Hulpke (2011). *Abstract Algebra in GAP*. Online version.

Introduction to Computer Programming - MTH00055

Module designation	Introduction to Computer Programming
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Nguyen Hien Luong MSc. Ha Van Thao
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: lectures 45 hours + 30 lab hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Introduce to students the logic of computer programming – how computers operate with programming languages and how to write instructions using flowchart or pseudocode; how to modularize a computer program; how to write computer programs in C language.
Content	Basic concepts of algorithms and algorithms. Describe the skills and fundamentals of computer programming. Introduction to the C programming language. Pointer. Array and string. Function. Structure.

Examination forms	Final exam: written exams.
Study and examination requirements	Test 1: 15% Test 2: 15% Test 3: 20% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Main: 1. Avelino J. Gonzalez, *Computer Programming in C for Beginners*, Springer Cham, 2020. References: 1. G. McKenzie. *Programming Languages: C and C++.* Murphy & Moore Publishing, 2022 2. Orhan Gazi. *Modern C Programming Including Standards C99, C11, C17, C23*. Springer Cham, 2023. 3. Pham Van At. *C and C# Programming Techniques: From Basics to Advanced.* Hanoi: Hanoi University of Science and Technology Publishing House, 2024.

II. PROFESSIONAL EDUCATION KNOWLEDGE

1. BASIC KNOWLEDGE

Algebra 2A - MTH10402

Module designation	Algebra 2A
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Tran Ngoc Hoi
Language	Vietnamese
Relation to curriculum	Compulsory for DS and MCS programmes
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030) • Linear Algebra Practice (MTH00083)
Module objectives/intended learning outcomes	General objective: To equip students with basic knowledge of advanced linear algebra, as a foundation for specialized courses in Mathematics. Specific objectives for course learning outcomes: Knowledge: Master the diagonalization and Jordanization of linear operators on finite-dimensional vector spaces, the structure of Euclidean spaces, bilinear forms, and quadratic forms. Skills: Diagonalization and Jordanization of matrices on numeric fields, computation in Euclidean space, quadratic transformations.

	Attitude, diligence: Serious, progressive.
Content	This course introduces the basic knowledge of linear operator reduction (diagonalization, Jordan canonical form), Euclidean space, bilinear form, and quadratic form.
Examination forms	Midterm and Final exam: written exam
Study and examination requirements	Tests - process score: 50% Final exam: 50% Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Bui, X.H., Tran, N.H., Le, V. L. (2017). *Linear Algebra and Its Applications*, Vol 2 (in Vietnamese). VNUHCM Press. 2. Ngo, V. T. (2001). *Textbook of Linear Algebra*. Vietnam National University Press, Hanoi. 3. David, C. L. (2011). *Linear Algebra and Its Applications*. Pearson Education.

Functional Analysis - MTH10403

Module designation	Functional Analysis
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Bui Le Trong Thanh
Language	Vietnamese
Relation to curriculum	Compulsory for AM and MCS programmes.
Teaching methods	lectures, practice sessions
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: lectures 45 hours + 30 exercise hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	General Objective: Functional analysis is where students get their first basic understanding of infinite-dimensional spaces. This knowledge is indispensable for many specializations in both theoretical and applied mathematics. This is one of the first courses where the ability to acquire and use abstract and precise mathematical reasoning is trained and tested. Specific objectives/course outcomes standards: Knowledge: understand and apply in specific situations some basic concepts and results about Euclidean spaces, l^p, L^p spaces, bounded functions spaces, space of continuous linear mappings, Hilbert space. Skills: systematically understanding the above objects, grasping the relationships between concepts and results. Distinguish and criticize arguments that do not meet the exact requirements of mathematics. Attitude, diligence: Seeing the need for generalized development, thereby helping to solve application problems, thereby having a serious, proactive, and self-disciplined learning attitude.
Content	Metric space, normed space, continuous linear mapping between two normed spaces and their fundamental theorems, Hilbert space. This subject is considered as the basic knowledge for university level Mathematics. An essential and useful course for mathematical models using function spaces and infinite dimensional spaces (in Algebra, Optimization, Probability-Statistics, Partial Differential Equations, Analysis, Mathematical Methods in Physics, Computer Science, ...).
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 50%; Final Exam: 50% Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Lecture Notes on Functional Analysis*, Dinh Ngoc Thanh and Huynh Quang Vu. 2. *Functional Analysis*, Duong Minh Duc, 2005. 3. *Functional Analysis*, Dinh Ngoc Thanh and Dang Duc Trong, 2011.

Measure Theory and Probability - MTH10401

Module designation	Measure Theory and Probability
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Prof. Dang Duc Trong
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Analysis A1 (MTH00010) • Calculus A1 (MTH00011)
Module objectives/intended learning outcomes	The objective of the module is to equip students with the basic knowledge of the foundation of measure theory and the theory of integral, and the application of the measure theory in probability theory.
Content	Elementary probability theory, abstract measure theory. Integration with respect to probability measures, expectation, and variance Random variables, Law of large numbers and limit theorems. Changes of measures and the Radon-Nikodym Theorem. Conditional expectations, filtrations, and martingales.
Examination forms	Midterm and final exam: written exams.

Study and examination requirements	Exercises 20%; Mid-term 30%; Final exam 50%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Trong, D. D., Thanh, D. N., Minh, N. D. (2024). *Probability Theory*. (in Vietnamese). VNUHCM Press. 2. Trong, D. D., Thanh, D. N. (2014). *Textbook of Measure Theory and Probability*, (in Vietnamese). VNUHCM Press. 3. K. L. Chung, *A Course in Probability Theory*, 3rd edition, Academic Press, 2001. 4. Quan, P. H., Thanh, D. N. (2011). *Probability and Statistics*. Vietnam Education Publishing House.

Mathematical Statistics - MTH10404

Module designation	Mathematical Statistics
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Dr. Hoang Van Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	lectures, exercises, practice sessions
Workload (incl. contact hours, self-study hours)	Total: 145 hours. Contact hours: Lectures: 15 hours (in class) Exercises: 30 hours (in class) Practice: 30 hours (in lab) Private study: 70 hours (self-study)
Credit points	3 Credits (5.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Probability and Statistics (MTH00040) - for students of DS programme.

	• Measure Theory and Probability (MTH10401) - for students of other programmes.
Module objectives/intended learning outcomes	This course provides foundations of statistical inference. Students will be able to make inferences from data, making decisions and predictions. Students can apply statistical techniques to practical problems using R or SPSS.
Content	Descriptive statistics. Sampling distributions. Point estimations: definition, estimators, and estimates. Unbiasedness, efficiency, consistency. Method of moments, method of maximum likelihood, Bayes estimators. Minimum variance unbiased estimator, Cramer-Rao lower bound. Confidence interval: CI for means, variances and proportions. Hypothesis testing: basic concepts, null and alternative hypotheses, simple and compound hypotheses, type I and II errors, critical region, size and power of a test, p-value. Neyman - Pearson lemma. Likelihood Ratio tests. z-tests and t-tests. Goodness of fit test. Linear Regression.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Homework exercises: 10% Computing exercises: 20% Mid-term: 20%. Final exam: 50%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Casella, George, and Roger L. Berger. *Statistical Inference*. Cengage Learning, 2021. 2. Lehmann, Erich Leo, Joseph P. Romano, and George Casella. *Testing Statistical Hypotheses*. Vol. 3. New York: Springer, 2005. 3. Montgomery, Douglas C., and George C. Runger. *Applied Statistics and Probability for Engineers*. John Wiley & Sons, 2010.

Data Structures and Algorithms - MTH10405

Module designation	Data Structures and Algorithms
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	MSc. Ha Van Thao
Language	Vietnamese
Relation to curriculum	Compulsory for DS and MCS programmes. Elective for others programme.
Teaching methods	lectures, exercises, practice sessions
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: lectures 45 hours + 30 practical lessons in the laboratory. Private study: 90 hours (self-study).
Credit points	4 credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Fundamentals of Computer Programming (MTH00055)
Module objectives/intended learning outcomes	Present basic knowledge of the data structure and algorithms, how to re-perform data according to problem purposes. Two basic methods of searching and eleven arrangements algorithms are presented in the second part. Chapter Three and Four will present the basic data structures. Specific objectives / course learning outcomes: Knowledge: Clearly understanding algorithms, search methods, arrangements, and dynamic data structures Skills: analyzing algorithms, generalize data, algorithm settings

	Attitudes, specialized: have attitudes, views, and proper perception of subjects
Content	Introduction of algorithms, analyzing the algorithms and generalizing data. Search methods and arrangement in arrays. Basic dynamic data structure: single and double linked lists, binary search trees.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Data Structures and Algorithms in Python 2nd edition*, Michael T. Goodrich, Roberto Tamassia, Michael Goldwasser, 2022, John Wiley & Sons Ltd. 2. *A Common-Sense Guide to Data Structures and Algorithms: Level Up Your Core Programming Skills*, Wengrow, Jay. - Raleigh, 2020, NC: The Pragmatic Bookshelf 3. *Data Structures and Algorithm Analysis in C++ 4th edition*, Mark Allen Weiss, 2021, Pearson

Discrete Mathematics - MTH10406

Module designation	Discrete Mathematics
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc. Ha Van Thao
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	lecture, exercise
Workload (incl. contact hours, self-study hours)	Total: 165 hours Contact hours:

	Lecture: 45 hours (in class) Exercise: 30 hours (in class) Private study: 90 hours (self-study).
Credit points	4 credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Fundamentals of Computer Programming (MTH00055)
Module objectives/intended learning outcomes	General objectives: providing basic knowledge for information technology students. Specific objectives / course learning outcome: 1. Knowledge: Mastering logical issues, set- mapping, relations on sets. 2. Skills: Strict and accurate presentation of professional issues, good implementation of counting problems, solution of recursion, integer algorithms, solving equations on Zn, minimal polynomial algorithms for Boole functions. 3. Attitude, diligence: serious and positive learning, participating in discussions, asking questions, and responding to comments, completing the lecturers' learning requirements.
Content	The subject is within scientific knowledge. It provides very necessary knowledge (about logic, discrete structures, related algorithms, ...) for information technology students. This knowledge helps students a lot to absorb their grassroots and specialized subjects.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nguyen Huu Anh. (2003). *Discrete Mathematics*. The Education Publishing House.

	2. Kenneth H. Rosen. (2019). *Discrete Mathematics and Its Application*. 8th ed. McGraw Hill. 3. Richard Johnsonbaugh. (2005). *Discrete Mathematics*. Pearson.

Object-Oriented Programming - MTH10407

Module designation	Object-Oriented Programming
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc. Nguyen Ngoc Long
Language	Vietnamese
Relation to curriculum	Compulsory for DS and MCS programmes. Elective for others programme.
Teaching methods	lecture, lab work
Workload (incl. contact hours, self-study hours)	Total: 165 hours Contact hours: lecture 45 hours + 30 exercise hours Private study: 90 hours (self-study).
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Data Structures and Algorithms (MTH10405) ● Fundamentals of Computer Programming (MTH00055)
Module objectives/intended learning outcomes	Introducing the basic principles of object-oriented methodology and object-oriented programming techniques as a basis for building later applications.
Content	The principles of object-oriented methodology. The principles for building object classes, data identification and manipulation, establishing relations between classes, especially inheritance relation and polymorphism.

	Design, build classes, definition operations and operations in C ++. Install specific relations between layers, inheritance, polymorphism in C ++.
Examination forms	Midterm and final exam: written exams/major assignment
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbooks: 1. Stroustrup, Bjarne. *Programming: Principles and Practice Using C++*. Addison-Wesley Professional, 2024. References: 1. Forouzan, Behrouz A., and Richard F. Gilberg. *C++ programming: an object-oriented approach*. McGraw-Hill, 2020. 2. Kirk, Dorothy. *Demystified Object-Oriented Programming with C+*. Packt Publishing, 2021. 3. Sundnes, Joakim. *Introduction to scientific programming with Python*. Springer Nature, 2020.

2. SPECIALIZATION IN DATA SCIENCE

2.1. COMPULSORY COURSES

Introduction to Database Systems - MTH10312

Module designation	Introduction to Database Systems
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc. Nguyen Hien Luong
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme; Compulsory for Specialization in Data Science and Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Introduce students to basic concepts of databases; data models; database design standards and standard forms.
Content	Basic concepts of databases. Linked entity data model, relational data model. Operations on the relational data model. SQL query language. Standard forms and normalization algorithms for databases.

Examination forms	Midterm and final exam: written exam
Study and examination requirements	Mid-term: 30%, Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	1. R. Elsmari, S. Navathe. *Fundamentals of Database Systems 7th edition*, Addison Wesley, 2016. 2. Hector Garcia-Molina, Jeffrey Ullman, Jennifer Widom. *Database Systems: The Complete Book*. Pearson, 2008. 3. S. K. Singh. *Database Systems: Concepts, Design and Applications*. Pearson, 2009.

Introduction to Artificial Intelligence - MTH10318

Module designation	Introduction to Artificial Intelligence
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Assoc. Prof. Nguyen Thanh Binh
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme; Compulsory for Specialization in Data Science and Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)

Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Data structures and Algorithms (MTH10405) ● Discrete Mathematics (MTH00041)
Module objectives/intended learning outcomes	The course provides some basic knowledge of artificial intelligence science. The main content including Problem solving methods and applications. Focus on heuristic methods. Presenting some knowledge representation methods and some knowledge processing techniques. The problem of approximate reasoning. Introduction to some knowledge systems. Introduce some other models and methods. Ontology and Agent Concepts. At the same time, the course also introduces soft computing techniques such as Fuzzy, Neural networks, and genetic algorithms.
Content	Overview of the science of AI State space and the search problem Knowledge demonstration Introduction to machine learning and knowledge discovery Introducing Ontology and Intelligent Agent Introduction to Genetic Algorithms (GA) Introducing Fuzzy Logic Neural Network
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Do theoretical and practical exercises (40%). Final exam (60%). Final score is greater or equal to 5.0/10.0
Reading list	1. *Artificial Intelligence: a Modern Approach 4th illustrated edition*, Stuart J. Russell, Peter Norvig, 2021, Pearson 2. *Artificial Intelligence: A Textbook*, Charu C. Aggarwal, 2021, Springer International Publishing

	3. *Artificial Intelligence with Python*, Teik Toe Teoh, Zheng Rong, 2022, Springer Nature Singapore

Introduction to Machine Learning - MTH10353

Module designation	Introduction to Machine Learning
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Huynh The Dang
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme; Compulsory for Specialization in Data Science and Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab works, project
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Probability and Statistics (MTH00040) ● Linear Algebra (MTH00030) ● Introduction to Artificial Intelligence (MTH10318) ● Python for Data Science (MTH10605)
Module objectives/intended learning outcomes	Students will be provided with basic knowledge of Machine Learning including decision trees, artificial neural networks, genetic algorithms, statistical theory of hypothesis testing

	and Bayes, ... From that students can learn more deeply about machine learning models such as association classifiers, clustering, Bayesian networks, ... This is necessary knowledge to serve the scientific research work of students in the future.
Content	1. Linear Regression 2. Logistic Regression 3. Decision Tree 4. Random Forest 5. K-Means Clustering 6. Support Vector Machine (SVM) 7. Artificial Neural Network (ANN)
Examination forms	Final exam: written exam
Study and examination requirements	Project: 15% Final exam: 75%.
Reading list	Main textbook: 1. Anil Ananthaswamy, *Why Machines Learn: The Elegant Math Behind Modern AI*, Dutton (Penguin Random House). 2024. 2. Laurence Moroney, *AI and Machine Learning for Coders: A Programmer's Guide to Artificial Intelligence*, O'Reilly Media. 2020. 3. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media. 2022. References: 1. tensorflow.org, TensorFlow Tutorials, https://www.tensorflow.org/tutorials, 2023. 2. keras.io, keras, https://keras.io/, 2023. 3. Scikit-learn.org, scikit-learn, https://scikit-learn.org/stable/tutorial/index.html, 2024

	4. Python https://www.python.org/, 13 June 2025. 5. PyTorch.org, PyTorch Tutorials, https://docs.pytorch.org/tutorials/, 2024

Data Mining - MTH10358

Module designation	Data Mining
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Nguyen Thanh Binh
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme. Compulsory for Specialization in Data Science, MCS programme. Elective for others programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Fundamentals of Computer Programming (MTH00055)
Module objectives/intended learning outcomes	After successfully completing this course, students will be able to: Analyze data and carry out the steps of the data mining process

	Understand and apply data mining algorithms and tools that can be used to assist data analysts and data mining application developers Explain common data mining tasks such as regression, classification, clustering, and association rule mining Participate in advanced research to improve existing algorithms for each specific problem in data mining.
Content	This course is intended to introduce the knowledge mining process, concepts, technologies, and applications of data mining. In addition, this course also covers data preprocessing problems, data mining tasks, algorithms, and data mining tools that can be used to assist data analysts and analysts. Data mining application development. Subject-specific topics including overview of data mining, data mining problems, data preprocessing problems, data regression, data classification, clustering data mining, association rule mining, data mining application development, and advanced research topics in data mining.
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Midterm: 30% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Han, Jiawei, Jian Pei, and Hanghang Tong. *Data mining: concepts and techniques.* Morgan kaufmann, 2022. References: 1. Ardilla, Yunita, et al. *Data Mining dan Aplikasinya*. Penerbit Widina, 2021. 2. Zaki, Mohammed J., Wagner Meira Jr, and Wagner Meira. *Data mining and machine learning: Fundamental concepts and algorithms*. Cambridge University Press, 2020.

	3. Witten, Ian H., et al. *Data Mining: Practical machine learning tools and techniques*. Elsevier, 2025.

Analysis of Statistical Data - MTH10513

Module designation	Analysis of Statistical Data
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. To Duc Khanh
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lectures, group work
Workload (incl. contact hours, self-study hours)	120 Hours Lectures: 30 hours (in class) Practice: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	3 Credits (4.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Measure Theory and Probability (MTH10401) ● Mathematical Statistics (MTH10404) Recommended prerequisites: ● Multivariate Statistics (MTH10619)
Module objectives/intended learning outcomes	The course provides students with a comprehensive perspective on data processing in general and statistical data analysis in particular. It equips students with the necessary skills to become data scientists, enabling them to independently complete an entire data processing cycle using statistical analysis methods with R programme. Knowledge: Gain a solid understanding of the steps involved in a statistical data processing cycle. Distinguish between different types of data and select appropriate approaches.

	Skills & competences: Through laboratory activities, the course provides the necessary tools for the data processing, using the R statistical software. By the laboratory activities and group work, the student will be able to: ● recognize and appropriately describe the case study. ● effectively apply various analytical methods to different types of data and organizational/business questions. ● increase sensitivity and criticality in the use of statistical methods with regard to case studies. ● work in a group. ● develop analytical skills and independent judgment. ● develop communication skills.
Content	This module includes the following topics: 1. Introduction to statistical data processing. 2. Data and some statistical concepts. 3. A/B testing. 4. Linear regressions and Prediction. 5. Classification models. 6. Strategy for missing data and imbalanced data. 7. Applied survival analysis to Data science
Examination forms	Midterm exam: written exams. Final exam: project.
Study and examination requirements	Class-attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	References: 1. Peter Bruce. & Andrew Bruce. *Practical Statistics for Data Scientists*, O'Reilly Media, Inc., 2017. 2. Gareth James, Daniela Witten, Trevor Hastie & Robert Tibshirani. *An Introduction to Statistical Learning with application in R, 2nd*, Springer, 2021.

	3. Alan Agresti & Maria Kateri. *Foundations of Statistics for Data Scientists*, Chapman & Hall/CRC, 2022. 4. Hadley Wickham, Garrett Grolemund. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*, O'Reilly Media, Inc., 2016.

Python for Data Science - MTH10605

Module designation	Python for Data Science
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Ha Van Thao
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme; Compulsory for Specialization in Data Science and Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Object Oriented Programming (MTH10407) ● Data Structure and Algorithm (MTH10405)
Module objectives/intended learning outcomes	General objective: To present basic knowledge of python programming for data science for multidimensional signal

	data processing in data classification or object recognition, data mining, data analysis, statistics, machine learning, …. The main content of the program will present the basis for students to be able to program in python in data science. Then students will apply the knowledge to solve large problems. Specific objectives/course output standards: Knowledge: Master the basics of python programming Skills: analyzing and applying algorithms in practice Attitude, diligence: have the right attitude, opinion, and awareness about the subject
Content	This module includes the following topics: Introduction to Python Programming Control structure Function Data structures in Python Object-oriented programming Support libraries
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Paul Deitel & Harvey Deitel, *Intro to Python for Computer Science and Data Science*, Pearson. 2022. 2. Anshuman Mishra, *Data Science with Python: A Complete Guide with Practical Examples and MCQs*

	(Python Programming), Amazon Digital Services (Kindle edition), 27 December 2024. 3. Jake VanderPlas, *Python Data Science Handbook: Essential Tools for Working with Data (2nd ed.)*, O'Reilly Media (Paperback/Kindle), 17 January 2023. References: 1. Rajeev Ratan, *Data Science & Deep Learning for Business*, Udemy, 2022. 2. tensorflow.org, TensorFlow Tutorials, https://www.tensorflow.org/tutorials, 2023. 3. keras.io, keras, https://keras.io/, 2023. 4. Scikit-learn.org, scikit-learn, https://scikit-learn.org/stable/tutorial/index.html, 2024 5. https://www.python.org/, 13 June 2025.

2.2. ELECTIVE COURSES

Pattern Recognition - MTH10322

Module designation	Pattern Recognition
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture and Lab work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours, Lab work: 30 hours

	Private study: 90 hours (Self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Computational Software Laboratory (MTH00087)
Module objectives/intended learning outcomes	This course provides fundamental knowledge of pattern recognition with examples from many applications, multidimensional data processing techniques, classification and regression algorithms as well as data clustering. Knowledge: familiarity with fundamental knowledge of pattern recognition such as multidimensional data processing techniques, classification algorithms, regression and data clustering Skills and competences: cognitive and practical abilities to use the knowledge in pattern recognition field, artificial neural network, SVM (support vector machine) and computer vision
Content	This module includes the following topics: 1. Basic concepts in pattern recognition 2. Bayesian discrimination principle 3. Parameter estimation 4. Non-parametric techniques 5. Linear discriminants 6. Support Vector Machine 7. Artificial Neural Network 8. Algorithm-independent machine learning 9. Unsupervised learning and clustering 10. Final Project
Examination forms	Class discussion; mid-term and final-term exam: written exam.
Study and examination requirements	Class attendance: 5%.

	Discussion, exercises, practices: 5%. Midterm exam: 20% Final theory exam: 70%. Final score is compulsory and greater or equal to 5.0/10.0.
Reading list	Main textbooks: 1. *Fundamentals of Pattern Recognition and Machine Learning*, U. Braga-Neto, Springer 2020. References: 1. *Pattern Recognition: Introduction, Features, Classifiers and Principles*, J. Beyerer, R. Hagmanns, D. Stadler, De Gruyter 2024. 2. *Pattern Recognition, An Introduction*, S. T. Ahmed, S. M. Basha, S. R. Arumugam, M. M. Kodabagi, MileStone Research Publications 2021.

Multidimensional Signal Processing - MTH10323

Module designation	Multidimensional Signal Processing
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Kha Tuan Minh
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)

Required and recommended prerequisites for joining the module	Required prerequisites: ● Linear Algebra (MTH00030) ● Measure Theory and Probability (MTH10401) ● Mathematical Statistics (MTH10404)
Module objectives/intended learning outcomes	General objective: Present some basic knowledge about multidimensional signal data processing in data classification or object recognition, data mining, data analysis, statistics, machine learning, etc. The main content of the program will present mathematical ideas of principal component analysis methods, discrete analysis, independent component analysis, non-negative matrices, and some multiplication methods. Students will then apply knowledge to solve major exercises. Specific objectives/subject output standards: 1. Knowledge: Master the basic knowledge of multidimensional processing 2. Skills: analyse and apply algorithms in practice 3. Attitude and diligence: have the right attitude, perspective and awareness about the subject
Content	1. Principal Component Analysis method 2. Discrete analysis 3. Independent Component Analysis method 4. Non-negative matrix factorization 5. Kernel methods 6. Clustering and classification
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5%

	Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Slides and lectures. 2. *Multidimensional Digital Signal Processing (Prentice-hall Signal Processing Series)* by Dan E. Dudgeon, Russell M. Mersereau References: 1. Richard O. Duda, Peter Elliot Hart, David G. Stork, *Pattern classification*, Wiley 2004. 2. Boris Mirkin, *Mathematical classification and clustering*, Khuwer Academic Publisher 1996. 3. Martin E. Modell, *Data analysis, data modelling, and classification*, New York: McGraw-Hill 1992.

Algorithm Analysis - MTH10325

Module designation	Algorithm Analysis
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Nguyen Thanh Binh
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Data Science and Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab work, practice in groups
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours:

	Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Discrete Mathematics (MTH10406)
Module objectives/intended learning outcomes	Equip students with knowledge and programming skills to evaluate the complexity of algorithms, correctness of algorithms, generating functions and applications; permutations and applications; evaluate some commonly used algorithms.
Content	1. Overview of the algorithm and its complexity 2. Generative functions and applications 3. Permutations and applications 4. Evaluate the complexity of some common algorithms
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Do theoretical and practical exercises in groups (10%). Midterm written exam (20%). Final written exam (70%). Final score is greater or equal to 5.0/10.0
Reading list	1. Susanna S. Epp, *Discrete Mathematics with Applications* (5th Edition), Cengage Learning, 2020. 2. Rajan Chattamvelli, Ramalingam Shanmugam, *Generating Functions in Engineering and the Applied Sciences*, Springer, 2023. 3. Irem Kucukoglu, Gradimir V. Milovanović & Yilmaz Simsek, *Analysis of Generating Functions for Special Words and Numbers*, ResearchGate, 2022.

	4. Henry Adams, Kelly Emmrich, Maria Gillespie, Shannon Golden, Rachel Pries, *Counting Rocks! An Introduction to Combinatorics*, arXiv, 2021. 5. *Combinatorial Algorithms*, Springer, IWOCA 2022 Proceedings, 2022. 6. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, *Introduction to Algorithms (4th Edition)*, The MIT Press, 2022. 7. Tim Roughgarden, *Algorithms Illuminated: Omnibus Edition*, Soundlikeyourself Publishing, 2022.

Software Testing - MTH10332

Module designation	Software Testing
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc. Nguyen Dat Thong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: To be familiar with software testing and software testing techniques.

	Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of software testing and software testing techniques, thereby understanding the importance of software testing in the software development process, understanding the role and work of testers. 2. Skills: research and work in the field of software testing. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 1. Overview of software testing 2. Basic concepts in software testing 3. Types of software testing 4. Software testing techniques 5. Project management in software testing
Examination forms	Class discussion; Midterm and final exam: written exams
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Hung Q. Nguyen, Bob Johnson, Michael Hackett and Robert Johnson, *Testing Web Applications*, VNUHCM Press, 2010. References: 1. Cem Kaner, Jack Falk and Hung Q. Nguyen, *Testing Computer Software*, 2nd Edition, Wiley, 1999.

	2. Bernard Homès. *Fundamentals of Software Testing*. Wiley-ISTE, 2012.

Software Testing 2 - MTH10336

Module designation	Software Testing 2
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Nguyen Dat Thong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Software Testing (MTH10332) ● Introduction to programming (MTH00055) ● Object Oriented Programming (MTH10407)
Module objectives/intended learning outcomes	General objectives: To be familiar with software test automation and techniques in automated software testing. Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of software test automation and techniques in automated software testing, thereby understanding the importance of automated software testing in the software

	development process; Understand the role and work of automation testing engineers. 2. Skills: research and work in the field of software testing automation. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 1. Overview of automated software testing 2. Automated unit testing with xUnit 3. Automated testing with Microsoft CodedUI 4. Automate testing of Web applications with Selenium 5. Automated performance testing with Microsoft Visual Studio
Examination forms	Class discussion; Final exam: written exams
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final practice exam: 50% Final theory exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Hung Q. Nguyen, Bob Johnson, Michael Hackett and Robert Johnson, *Testing Web Applications*, VNUHCM Press, 2010. References: 1. Cem Kaner, Jack Falk and Hung Q. Nguyen, *Testing Computer Software*, 2nd Edition, Wiley, 1999. 2. Selenium HQ: http://docs.seleniumhq.org/docs/ 3. JUnit: http://junit.org/junit4/ 4. NUnit: https://www.nunit.org/index.php?p=documentation

	5. CodedUI Test: https://msdn.microsoft.com/en-us/library/dd286726.aspx

Database Management System - MTH10344

Module designation	Database Management System
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Tran Anh Tuan MSc. Nguyen Hien Luong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	● Introduction to Database Systems (MTH10312)
Module objectives/intended learning outcomes	Students know how to access knowledge about databases and database management systems: components of the SQL Server database management system and their functions, concurrent access management mechanisms, safety and data recovery after incidents, decentralisation and security.
Content	Introduction including definition, functions, properties, architecture of database management system, SQL and Microsoft SQL server Fundamental SQL syntaxes Securing SQL server

	Stored and manage procedures Creating and managing user-defined functions Creating and managing triggers in SQL server
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	1. Duong Quang Thien. *SQL Server 2000: T-SQL Programming.* Ho Chi Minh City: Saigon Culture Publishing House, 2007. 2. Ray Rankins, Paul Bertucci, Chris Gallelli, Alex T. Silverstein, *Microsoft SQL Server 2005 Unleashed*, Sams Publishing, 2007. 3. Brian Knightet al, *Professional SQL Server 2005 Administration*, Wrox Press, 2007. 4. Paul Turley & Dan Wood, *Beginning Transact-SQL with SQL Server 2000 and 2005*, Wrox Press, 2006.

Advanced Machine Learning - MTH10354

Module designation	Advance Machine Learning
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assoc. Prof. Pham The Bao
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture and lab work

Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours, Lab work: 30 hours Private study: 90 hours (Self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Introduction to artificial intelligence (MTH10318)
Module objectives/intended learning outcomes	This course provides advanced knowledge of machine learning. Knowledge: mastering the advanced knowledge of machine learning. Skills: cognitive and practical abilities to use this knowledge in practical problems. Attitude: Have the right attitude, perspective and awareness about the subject.
Content	This module includes the following topics: Depth first, Breadth first, complexity, completeness and optimality of search methods, Implementing DFS and BFS, Iterative deepening search Using heuristics for search, hill-climbing, best-first, beam search Optimal paths, Branch and Bound, A* Parallel search, Bi-directional search Games, minimax, Alpha-beta pruning Constraint satisfaction search, Cryptographic problems, Real-time A*, Iterative-deepening A* Knowledge Representation and Reasoning: Building a Knowledge Base: Propositional logic, Predicate logic, Theorem Proving.

Examination forms	Class discussion; mid-term and final term exam: written exam.
Study and examination requirements	Class - attendance: 5%. Discussion, exercises, practices: 5%. Midterm exam: 20% Final theory exam: 70%. Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Buduma, Nithin, Nikhil Buduma, and Joe Papa. *Fundamentals of deep learning: Designing next-generation machine intelligence algorithms*. " O'Reilly Media, Inc.", 2022. Reference: 1. Zaki, Mohammed J., Wagner Meira Jr, and Wagner Meira. *Data mining and machine learning: Fundamental concepts and algorithms*. Cambridge University Press, 2020. 2. Jung, Alexander. *Machine learning: the basics*. Springer Nature, 2022.

Natural Language Processing - MTH10359

Module designation	Natural Language Processing
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Huynh Thanh Son
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	lecture, lesson, lab work, project.
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours:

	Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Introduction to Machine Learning (MTH10353) ● Introduction to Artificial Intelligence (MTH10318)
Module objectives/intended learning outcomes	Equip students with the basic knowledge of natural language processing (NLP), the methods and techniques used in NLP, along with fundamental mathematical models applied to text processing tasks in non-random contexts.
Content	Including the theory of natural language processing (NLP) models, syntactic and semantic analysis, language generation, and the impact of computational methods and algorithms in NLP applications.
Examination forms	lab work and oral presentation
Study and examination requirements	lab works: 30%, oral presentation: 70%
Reading list	Main textbooks: 1. *Natural Language Processing in Action: Understanding, Analyzing, and Generating Text with Python* (2019) References: 1. *Practical Natural Language Processing* — O'Reilly (2020) 2. *Natural Language Processing with Transformers* — O'Reilly (2022) 3. *Transformers for Natural Language Processing: Build Innovative Deep Neural Network Architectures for NLP with Python, PyTorch, TensorFlow, BERT, RoBERTa, and More* (2021)

Optimization Algorithms - MTH10450

Module designation	Optimization Algorithms
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Vo Si Trong Long
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Optimization, AM programme. Elective for others programme.
Teaching methods	Lecture, lab work, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 lectures hours + 30 lab works hours (in class) Private study: 90 hours (self-study)
Credit points	4 credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Provide students with knowledge and methods to solve constrained and unconstrained optimization problems.
Content	1. Basic properties and concepts of algorithms, Direct/Indirect line search method for unconstrained optimization problem. 2. Deep reduction method, gradient method and conjugate gradient method.

	3. Newton and quasi-Newton methods. 4. The least squares problem. Kuhn-Tucker Optimality Conditions.
Examination forms	Mid-term and Final exam: Written exam.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nocedal J, Wright SJ: *Numerical optimization*. Springer, New York, 1999 2. Polak Elijah, *Optimization: algorithms and consistent approximations*, Springer, 1997. 3. Alaa Khamis (2024). *Optimization Algorithms*. Manning Publications.

Numerical Methods for Data Science - MTH10607

Module designation	Numerical Methods for Data Science
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Nguyen Thi Hoai Thuong.
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objective: Present numerical methods for processing multidimensional signal data in data classification or object recognition, data mining, data analysis, statistics, machine learning, etc. The main content of the program will present mathematical ideas of numerical methods in data science. Students will then apply knowledge to solve major exercises. Specific objectives/subject output standards: 1. Knowledge: Master the basic knowledge of numerical methods in data science 2. Skills: analyse and apply algorithms in practice 3. Attitude and diligence: have the right attitude, perspective and awareness about the subject.
Content	1. Learning theory 2. Linearity 3. Multiplicative weights and online learning 4. Optimization 5. Regression and its analysis 6. Graphical Models 7. Algorithms for massive data sets
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	1. Boyd S. Vandenberghe L., *Convex Optimization*, Cambridge 2009.

	2. Avrim Blum, John Hopcroft, Ravindran Kannan, *Foundations of Data Science*, Cambridge University Press 2016. 3. Rolf Wuthrich, Carole El Ayoubi. *Numerical Methods for Engineering and Data Science*. CRC Press 2025.

Data Visualization - MTH10608

Module designation	Data Visualization
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. To Duc Khanh
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Data Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab work, group work
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Measure Theory and Probability (MTH10401) ● Mathematical Statistics (MTH10404) Recommended prerequisites: ● Multivariate Statistics (MTH10619)

Module objectives/intended learning outcomes	The course provides students with a comprehensive understanding of data visualization. It equips them with the skills to identify data types and select appropriate methods and tools for visually representing data in a clear and intuitive manner, maximizing support for further analytical insights. Additionally, students will develop proficiency in using Power BI for data visualization. Knowledge: Gain a solid understanding of the fundamental principles of data visualization and the various types of charts used in the process. Skills & competences: Apply core principles effectively to create suitable visual representations. Develop the ability to choose the right chart for different data types and analytical purposes. Gain proficiency in using Power BI to create visualizations. Through laboratory activities and group work, the student will be able to: ● recognize and appropriately describe the case study; ● identify the appropriate visualiazion methods; ● teamwork; ● develop visualiazion skills and independent judgment; ● develop communication skills.
Content	This module includes the following topics: 1. Introduction to data visualization (coordinate systems and axes, color scales). 2. Visualizing a single variable (quantitative and qualitative variable: amount, proportion). 3. Visualizing multiple variables (scatterplots, correlograms, dimension reduction). 4. Visualizing trends (smoothing, time-series). 5. Visualizing uncertainty (distribution, confidence regions).
Examination forms	Midterm exam: written exams.

	Final exam: project.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	References: 1. Claus O. Wilke. *Fundamentals of Data Visualization*, O'Reilly Media, 2019. 2. Knaflic, Cole Nussbaumer. *Storytelling with Data: A Data Visualization Guide for Business Professionals*, 2015. 3. Devin Knight, Erin Ostrowsky, Mitchell Pearson, Bradley Schacht. *Microsoft Power BI Quick Start Guide: The Ultimate Beginner's Guide to Data Modeling, Visualization, Digital Storytelling, and More*. Packt Publishing Limited, 2022.

Multivariate Statistics - MTH10619

Module designation	Multivariate Statistics
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Nguyen Thi Mong Ngoc
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, do exercises, group work, lab works
Workload (incl. contact hours, self-study hours)	Total: 165 hours Contact hours: Lectures 45 hours (in class) Practice: 30 hours (in lab) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)

Required and recommended prerequisites for joining the module	Required prerequisites: ● Linear Algebra (MTH00034) ● Algebra 2A (MTH10441) ● Mathematical Statistics (MTH10131)
Module objectives/intended learning outcomes	Equip students with the knowledge base of multidimensional statistical system processing. Apply multivariate skills and "hands-on" techniques using R or Python programming in analyzing the real data.
Content	Multivariate normal distribution, Inference about a mean vector (Hotelling's T^2 and Likelihood Ratio Tests, Confidence regions and simultaneous comparisons of Component Means, multivariate quality Control Charts, …); comparisons of several multivariate means (Comparing Mean Vectors from two population, One-Way MANOVA, Two-Way MANOVA, Testing for Equality of Covariance Matrices), Principal Components Analysis (PCA). Apply multivariate skills and "hands-on" techniques using R (or Python) programming in analyzing the real data.
Examination forms	Written exams
Study and examination requirements	Mid-term: 50%, Final exam: 50%.
Reading list	1. *Applied Multivariate Statistical Analysis*, Richard A. Johnson, Dean W. Wichern, 2016. 2. *An Introduction to Multivariate Statistical Analysis*, T. W. Anderson, 2003. 3. *Applied Multivariate StatisticalAnalysis*, Wolfgang Härdle, Léopold Sima, 2007. 4. *Applied Multivariate Statistics with R*, Daniel Zelterman, 2015. 5. *An R and S-PLUS Companion to Multivariate Analysis*, Everitt, B.S. 2005.

Seminar on Data Science - MTH10620

Module designation	Seminar on Data Science
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assigned lecturers of the Faculty of Mathematics and Computer Science, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Seminar, project
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Discussion: 60 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Learn more deeply about your major with greater self-study under the guidance of a lecturer and prepare to write a graduation thesis. Specific objectives/subject output standards: Knowledge: provides additional knowledge that is more extensive than the subjects in the major. Skills: prepare skills for deeper learning and research, such as: self-study, group discussion, writing reports/presentations, arguing/justifying/defending opinions, how to find and use resources related references.
Content	Students will work directly with faculty on a topic. Topic content can be: Issues that have not been mentioned in the subjects of the major.

	Problems arise in specialised subjects but have not been resolved. The necessary knowledge is not included in the program to prepare for the expected graduation thesis topic. The problems of the subject will be assigned by the lecturer for students to explore, research and present. Applicable forms: Under the guidance of instructors, students choose a topic and register for research tasks, then report back. Instructors analyse and evaluate results. The lecturer selects and presents the problem. Students present their understanding of that issue.
Examination forms	Final exam: project or representation.
Study and examination requirements	Class-attendance: 15% Discussion and practices: 35% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Depends on the assigned topics.

3. SPECIALIZATION IN MATHEMATICAL METHODS IN COMPUTER SCIENCE

3.1. COMPULSORY COURSES

Image Processing and Analysis - MTH10317

Module designation	Image Processing and Analysis
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Lab Works: 30 hours (in class) Practice: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Data Structures and Algorithms (MTH10405)
Module objectives/intended learning outcomes	General objectives: to be familiar with image processing and analysis. Specific objectives/course learning outcomes: 1. Knowledge: Presenting from the most basic concepts of image processing to processing methods in gray and color images, such as noise filtering methods, edge highlighting, morphological operations, wavelets, image segmentation, find edges and polygons and describe them. And the methods of

	image compression as well as basic watermarking techniques. 2. Skills: research and work in the field of image processing and analysis. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 1. An overview and basic knowledge of digital images 2. Intensity Transformation and Filtering 3. Filtering in the frequency domain 4. Restoration and reconstruction of image 5. Color image processing 6. Wavelet transform and multi-resolution processing 7. Image Compression 8. Image processing using morphological math 9. Image Segmentation 10. Performance and description
Examination forms	Class discussion; Seminar; Final exam: written exams
Study and examination requirements	Do theoretical and practical exercises in groups (30%) Seminar (20%) Final exam (50%) Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. A. Torralba, P. Isola, and W.T. Freeman. *Foundations of Computer Vision*. MIT Press, 2024. References: 1. S.S. Rajput, N.U. Khan, A.K. Singh, and K.V. Arya. *Digital Image Enhancement and Reconstruction*. Elsevier Science, 2022.

	2. Ryan Gillard Valliappa Lakshmanan Martin G¨orner. *Practical Machine Learning for Computer Vision: End-to-End Machine Learning for Images*. O'Reilly Media, Inc., 2021 3. Caide Xiao. *Mastering Computer Vision with PyTorch and Machine Learning*. IOP Publishing, 2024

Introduction to Artificial Intelligence - MTH10318

Module designation	Introduction to Artificial Intelligence
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Assoc. Prof. Nguyen Thanh Binh
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme; Compulsory for Specialization in Data Science and Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Data structures and Algorithms (MTH10405) ● Discrete Mathematics (MTH00041)

Module objectives/intended learning outcomes	The course provides some basic knowledge of artificial intelligence science. The main content including Problem solving methods and applications. Focus on heuristic methods. Presenting some knowledge representation methods and some knowledge processing techniques. The problem of approximate reasoning. Introduction to some knowledge systems. Introduce some other models and methods. Ontology and Agent Concepts. At the same time, the course also introduces soft computing techniques such as Fuzzy, Neural networks, genetic algorithms.
Content	Overview of the science of AI State space and the search problem Knowledge demonstration Introduction to machine learning and knowledge discovery Introducing Ontology and Intelligent Agent Introduction to Genetic Algorithms (GA) Introducing Fuzzy Logic Neural Network
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Do theoretical and practical exercises (40%). Final exam (60%). Final score is greater or equal to 5.0/10.0
Reading list	1. *Artificial Intelligence A Modern Approach 4th illustrated edition*, Stuart J. Russell, Peter Norvig, 2021, Pearson 2. *Artificial Intelligence: A Textbook*, Charu C. Aggarwal, 2021, Springer International Publishing 3. *Artificial Intelligence with Python*, Teik Toe Teoh, Zheng Rong, 2022, Springer Nature Singapore

Introduction to Cryptography - MTH10319

Module designation	Introduction to Cryptography

Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites ● Linear Algebra A1 (MTH00030) ● Abstract Algebra (MTH00031) ● Arithmetic and Algorithms (MTH10324)
Module objectives/intended learning outcomes	Equip students with basic knowledge about classical cryptographic theory, modern symmetric cryptography, public cryptography, cryptographic hash function, digital signature, key exchange protocol. At the same time, students are strengthened: programming skills through exercises on installing algorithms; skills in research and presentation of open topics; skills in using security software.
Content	1. Basic concepts of classical cryptographic theory. 2. Common symmetric cryptosystems: DES, AES. 3. Common cryptographic hash functions: MD5, SHA1. 4. Common public cryptosystems: RSA, ElGamal, ECC.

	5. Diffie-Hellman key exchange protocol 6. Digital signatures: RSA, DSA, ECC. 7. Using software: PGP, SSH, Cryptool
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	1. Douglas R. Stinson, Cryptography: *Theory and Practice*, 3rd. ed., Chapman & Hall/CRC, 2006. 2. Alfred J. Menezes, Paul C. van Oorsczyk and Scott A. Vanstone, *Handbook of Applied Cryptography*, CRC Press, 2001. 3. Pham Huy Dien, Ha Huy Khoai, *Information Coding: Mathematical foundations and applications*, National University Publishing House, Hanoi, 2003.

Digital Signal Processing - MTH10320

Module designation	Digital Signal Processing
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Nguyen Ngoc Long
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture

Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Introduction to the mathematical basis of digital signal processing, frequency representation of a signal as a linear combination of basic functions, classical (Fourier) and modern (wavelet) frequency representations.
Content	This module includes the following topics: 1. Frequency analysis of signals – general theory. 2. Fourier analysis. 3. Wavelet expansion.
Examination forms	Seminar; Final exam: written exams
Study and examination requirements	Do theoretical and practical exercises in groups (30%). Seminar (20%). Final exam (50%). Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Zieliński, T. P. (2021). *Starting Digital Signal Processing in Telecommunication Engineering: A Laboratory-based Course*. Germany: Springer International Publishing. References:

	2. Khan, Muhammad, et al. *Digital Signal Processing: A Breadth-First Approach*. Denmark, River Publishers, 2022. 3. Emily McLaughlin, *Advances in Digital Signal Processing*. United States, Murphy & Moore Publishing, 2022. 4. Anand, R. . *Digital Signal Processing: An Introduction*. Germany: Mercury Learning and Information., 2022.

High Performance Computing - MTH10321

Module designation	High Performance Computing
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assoc. Prof. Nguyen Thanh Binh
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab work
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Data Structures and Algorithms (MTH10406) ● Discrete Mathematics (MTH10406)

	● Object-Oriented Programming (MTH10407)
Module objectives/intended learning outcomes	Presents some basic knowledge in parallel programming models, parallel computer architecture, parallel programming models, and parallel computing algorithms.
Content	1. Fundamental definition 2. Parallelism methods 3. Message Passing Interface 4. Parallel programming schemes 5. Parallel algorithm
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Alsuwaiyel, M. H. Parallel Algorithms. Singapore: World Scientific Publishing Company, 2022 References: 1. Kumar, Subodh. *Introduction to Parallel Programming*. United States, Cambridge University Press, 2022. 2. Nelli, Fabio. *Parallel and High-Performance Programming with Python: Unlock Parallel and Concurrent Programming in Python Using Multithreading, CUDA, Pytorch, and Dask*. India, Orange Education Pvt Limited, 2023. 3. *Parallel and Distributed Computing, Applications and Technologies: 21st International Conference*, PDCAT 2020, Shenzhen, China, December 28–30,

	2020, Proceedings. Germany, Springer International Publishing, 2021.

Arithmetic and Algorithms - MTH10324

Module designation	Arithmetic and Algorithms
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Tran Nam Dung
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, project, seminar
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Abstract Algebra (MTH00031)
Module objectives/intended learning outcomes	Equip students with basic knowledge and programming skills of algorithmic arithmetic: prime numbers, congruences, continuous fractions, arithmetic functions, squared remainders, elliptical curves, algorithms deterministic and probabilistic primes testing, integer analysis, and discrete logarithms.
Content	1. Recall the basics of algorithms and complexity.

	2. Integer representation, prime numbers, Euclidean division algorithm, Chinese remainder theorem, Fermat's theorem, and continuous fractions 3. Euler nonfunctions, Mersenne primes, primitive roots 4. Squared residuals, Legendre notation, Jacobi notation 5. Prime number checking algorithms, prime number analysis 6. Elliptic curves over real numbers and over finite fields 7. Apply arithmetic to cryptographic theory: Caesar cipher system, RSA, ElGamal. 8. Algorithms to calculate discrete logarithms.
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Class attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbook: 1. Landau, Edmund. *Elementary number theory*. Vol. 125. American Mathematical Society, 2021. References: 1. Hasse, Helmut. *Number theory*. Walter de Gruyter GmbH & Co KG, 2022. 2. Baker, Alan. *Transcendental number theory*. Cambridge university press, 2022. 3. Aumasson, Jean-Philippe. *Serious cryptography: a practical introduction to modern encryption*. No Starch Press, Inc, 2024.

	4. Easttom, William. *Modern cryptography: applied mathematics for encryption and information security*. Springer Nature, 2022.

Algorithm Analysis - MTH10325

Module designation	Algorithm Analysis
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Nguyen Thanh Binh
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Data Science and Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab work, practice in groups
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Discrete Mathematics (MTH10406)
Module objectives/intended learning outcomes	Equip students with knowledge and programming skills to evaluate the complexity of algorithms, correctness of algorithms, generating functions and applications; permutations and applications; evaluate some commonly used algorithms.

Content	1. Overview of the algorithm and its complexity 2. Generative functions and applications 3. Permutations and applications 4. Evaluate the complexity of some common algorithms
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Do theoretical and practical exercises in groups (10%). Midterm written exam (20%). Final written exam (70%). Final score is greater or equal to 5.0/10.0
Reading list	1. Susanna S. Epp, *Discrete Mathematics with Applications* (5th Edition), Cengage Learning, 2020. 2. Rajan Chattamvelli, Ramalingam Shanmugam, *Generating Functions in Engineering and the Applied Sciences*, Springer, 2023. 3. Irem Kucukoglu, Gradimir V. Milovanović & Yilmaz Simsek, *Analysis of Generating Functions for Special Words and Numbers*, ResearchGate, 2022. 4. Henry Adams, Kelly Emmrich, Maria Gillespie, Shannon Golden, Rachel Pries, *Counting Rocks! An Introduction to Combinatorics*, arXiv, 2021. 5. *Combinatorial Algorithms*, Springer, IWOCA 2022 Proceedings, 2022. 6. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, *Introduction to Algorithms (4th Edition)*, The MIT Press, 2022. 7. Tim Roughgarden, *Algorithms Illuminated: Omnibus Edition*, Soundlikeyourself Publishing, 2022.

3.2. ELECTIVE COURSES

Pattern Recognition - MTH10322

Module designation	Pattern Recognition

Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture and Lab work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours, Lab work: 30 hours Private study: 90 hours (Self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Computational Software Laboratory (MTH00087)
Module objectives/intended learning outcomes	This course provides fundamental knowledge of pattern recognition with examples from many applications, multidimensional data processing techniques, classification and regression algorithms as well as data clustering. Knowledge: familiarity with fundamental knowledge of pattern recognition such as multidimensional data processing techniques, classification algorithms, regression and data clustering Skills and competences: cognitive and practical abilities to use the knowledge in pattern recognition field, artificial neural network, SVM (support vector machine) and computer vision
Content	This module includes the following topics: 11. Basic concepts in pattern recognition 12. Bayesian discrimination principle 13. Parameter estimation

	14. Non-parametric techniques 15. Linear discriminants 16. Support Vector Machine 17. Artificial Neural Network 18. Algorithm-independent machine learning 19. Unsupervised learning and clustering 20. Final Project
Examination forms	Class discussion; mid-term and final-term exam: written exam.
Study and examination requirements	Class attendance: 5%. Discussion, exercises, practices: 5%. Midterm exam: 20% Final theory exam: 70%. Final score is compulsory and greater or equal to 5.0/10.0.
Reading list	Main books: 1. *Fundamentals of Pattern Recognition and Machine Learning*, U. Braga-Neto, Springer 2020. References: 1. *Pattern Recognition: Introduction, Features, Classifiers and Principles*, J. Beyerer, R. Hagmanns, D. Stadler, De Gruyter 2024. 2. *Pattern Recognition, An Introduction*, S. T. Ahmed, S. M. Basha, S. R. Arumugam, M. M. Kodabagi, MileStone Research Publications 2021.

Software Testing - MTH10332

Module designation	Software Testing
Semester(s) in which the module is taught	4th semester

Person responsible for the module	MSc. Nguyen Dat Thong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: To be familiar with software testing and software testing techniques. Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of software testing and software testing techniques, thereby understanding the importance of software testing in the software development process, understanding the role and work of testers. 2. Skills: research and work in the field of software testing. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 6. Overview of software testing 7. Basic concepts in software testing 8. Types of software testing 9. Software testing techniques 10. Project management in software testing

Examination forms	Class discussion; Midterm and final exam: written exams
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Hung Q. Nguyen, Bob Johnson, Michael Hackett and Robert Johnson, *Testing Web Applications*, VNUHCM Press, 2010. References: 1. Cem Kaner, Jack Falk and Hung Q. Nguyen, *Testing Computer Software*, 2nd Edition, Wiley, 1999. 2. Bernard Homès. *Fundamentals of Software Testing*. Wiley-ISTE, 2012.

Software Testing 2 - MTH10336

Module designation	Software Testing 2
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Nguyen Dat Thong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class)

	Private study: 75 hours (self-study)
Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Software Testing (MTH10332) ● Introduction to programming (MTH00055) ● Object Oriented Programming (MTH10407)
Module objectives/intended learning outcomes	General objectives: To be familiar with software test automation and techniques in automated software testing. Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of software test automation and techniques in automated software testing, thereby understanding the importance of automated software testing in the software development process; Understand the role and work of automation testing engineers. 2. Skills: research and work in the field of software testing automation. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 6. Overview of automated software testing 7. Automated unit testing with xUnit 8. Automated testing with Microsoft CodedUI 9. Automate testing of Web applications with Selenium 10. Automated performance testing with Microsoft Visual Studio
Examination forms	Class discussion; Final exam: written exams
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final practice exam: 50% Final theory exam: 50%

	Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Hung Q. Nguyen, Bob Johnson, Michael Hackett and Robert Johnson, *Testing Web Applications*, VNUHCM Press, 2010. References: 1. Cem Kaner, Jack Falk and Hung Q. Nguyen, *Testing Computer Software*, 2nd Edition, Wiley, 1999. 2. Selenium HQ: http://docs.seleniumhq.org/docs/ 3. JUnit: http://junit.org/junit4/ 4. NUnit: https://www.nunit.org/index.php?p=documentation 5. CodedUI Test: https://msdn.microsoft.com/en-us/library/dd286726.aspx

Seminar on Computer Science - MTH10346

Module designation	Seminar on Computer Science
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assigned lecturers of the Faculty of Mathematics and Computer Science, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Seminar, project
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Discussion: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)

Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Learn more deeply about your major with greater self-study under the guidance of a lecturer and prepare to write a graduation thesis. Specific objectives/subject output standards: 1. Knowledge: provides additional knowledge that is more extensive than the subjects in the major. 2. Skills: prepare skills for deeper learning and research, such as: self-study, group discussion, writing reports/presentations, arguing/justifying/defending opinions, how to find and use resources related references.
Content	Students will work directly with faculty on a topic. Topic content can be: 1. Issues that have not been mentioned in the subjects of the major. 2. Problems arise in specialised subjects but have not been resolved. 3. The necessary knowledge is not included in the program to prepare for the expected graduation thesis topic. 4. The problems of the subject will be assigned by the lecturer for students to explore, research and present. Applicable forms: 5. Under the guidance of instructors, students choose a topic and register for research tasks, then report back. Instructors analyse and evaluate results. 6. The lecturer selects and presents the problem. Students present their understanding of that issue.
Examination forms	Final exam: project or representation.

Study and examination requirements	Class-attendance: 15% Discussion and practices: 35% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Depending on the assigned topics.

Information Theory - MTH10347

Module designation	Information Theory
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Bui Van Thach
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lectures, lab work, problem sets, seminar
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Mathematical maturity and a working knowledge of probability theory. Required prerequisites: • Measure Theory and Probability (MTH10401)
Module objectives/intended learning outcomes	Understand and quantify the limits of compressing and reliably storing/communicating data. Understanding how to apply information theory in some specific areas of science and engineering
Content	1. Introduction.

	2. Probability theory. 3. Information measures. 4. Symbol coding and compression. 5. Block coding. 6. Channel coding.
Examination forms	Problem sets and take-home exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade to pass at least 5.0/10.0.
Reading list	1. Cover, Thomas M. *Elements of information theory*. John Wiley & Sons, 1999. 2. MacKay, David JC. *Information theory, inference and learning algorithms*. Cambridge university press, 2003. 3. Gallager, Robert G. Information theory and reliable communication. Vol. 588. New York: Wiley, 1968.

Introduction to Machine Learning - MTH10353

Module designation	Introduction to Machine Learning
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Huynh The Dang
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme; Compulsory for Specialization in Data Science and Mathematical Methods in Computer Science, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab works, project
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class)

	Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Probability and Statistics (MTH00040) ● Linear Algebra (MTH00030) ● Introduction to Artificial Intelligence (MTH10318) ● Python for Data Science (MTH10605)
Module objectives/intended learning outcomes	Students will be provided with basic knowledge of Machine Learning including decision trees, artificial neural networks, genetic algorithms, statistical theory of hypothesis testing and Bayes, ... From that students can learn more deeply about machine learning models such as association classifiers, clustering, Bayesian networks, ... This is necessary knowledge to serve the scientific research work of students in the future.
Content	1. Linear Regression 2. Logistic Regression 3. Decision Tree 4. Random Forest 5. K-Means Clustering 6. Support Vector Machine (SVM) 7. Artificial Neural Network (ANN)
Examination forms	Final exam: written exam
Study and examination requirements	Project: 15% Final exam: 75%.
Reading list	Main textbooks: 1. Anil Ananthaswamy, *Why Machines Learn: The Elegant Math Behind Modern AI*, Dutton (Penguin Random House). 2024.

	2. Laurence Moroney, *AI and Machine Learning for Coders: A Programmer's Guide to Artificial Intelligence*, O'Reilly Media. 2020. 3. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media. 2022. References: 1. tensorflow.org, TensorFlow Tutorials, https://www.tensorflow.org/tutorials, 2023. 2. keras.io, keras, https://keras.io/, 2023. 3. Scikit-learn.org, scikit-learn, https://scikit-learn.org/stable/tutorial/index.html, 2024 4. Python https://www.python.org/, 13 June 2025. 5. PyTorch.org, PyTorch Tutorials, https://docs.pytorch.org/tutorials/, 2024

Advanced Machine Learning - MTH10354

Module designation	Advance Machine Learning
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assoc. Prof. Pham The Bao
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture and lab work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours, Lab work: 30 hours Private study: 90 hours (Self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites:

	• Introduction to Artificial Intelligence (MTH10318)
Module objectives/intended learning outcomes	This course provides advanced knowledge of machine learning. Knowledge: mastering the advanced knowledge of machine learning. Skills: cognitive and practical abilities to use this knowledge in practical problems. Attitude: Have the right attitude, perspective and awareness about the subject.
Content	This module includes the following topics: Depth first, Breadth first, complexity, completeness and optimality of search methods, Implementing DFS and BFS, Iterative deepening search Using heuristics for search, hill-climbing, best-first, beam search Optimal paths, Branch and Bound, A* Parallel search, Bi-directional search Games, minimax, Alpha-beta pruning Constraint satisfaction search, Cryptographic problems, Real-time A*, Iterative-deepening A* Knowledge Representation and Reasoning: Building a Knowledge Base: Propositional logic, Predicate logic, Theorem Proving.
Examination forms	Class discussion; mid-term and final term exam: written exam.
Study and examination requirements	Class - attendance: 5%. Discussion, exercises, practices: 5%. Midterm exam: 20% Final theory exam: 70%. Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks:

	1. Buduma, Nithin, Nikhil Buduma, and Joe Papa. *Fundamentals of deep learning: Designing next-generation machine intelligence algorithms*. " O'Reilly Media, Inc.", 2022. Reference: 1. Zaki, Mohammed J., Wagner Meira Jr, and Wagner Meira. *Data mining and machine learning: Fundamental concepts and algorithms*. Cambridge University Press, 2020. 2. Jung, Alexander. *Machine learning: the basics*. Springer Nature, 2022.

Mobile Computing - MTH10355

Module designation	Mobile Computing
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture and lab work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours, Labwork: 30 hours Private study: 90 hours (Self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	For this course, the modern world of computing is defined as one based on mobile devices (e.g., smartphones, netbooks, tablets, and similar) as the fundamental computing units that rely on cloud storage (e.g., Google Apps, Flickr/Picasa,

	Facebook, file backup systems such as Carbonite, and similar). In recent years, new, small and lightweight, but more intelligent mobile devices and embedded systems, such as smartphones, tablet devices, and other mobile Internet devices, have become popular. These devices present new challenges in programming, security, and data management. This course will introduce the basic concepts and issues in programming mobile devices; mobile device architecture; infrastructures needed to support such devices; power management issues; and information security, data management, and privacy issues. Specific objectives/course learning outcomes: 1. Knowledge: get familiar with knowledge of mobile computing 2. Skills: cognitive and practical abilities to use this knowledge in practical problems. 3. Attitude: Have the right attitude, perspective and awareness about the course.
Content	This module includes the following topics: 1. Section 1: What is Mobile Computing? An Overview 2. Section 2: Wireless Generations: An Overview 3. Section 3: Mobile Communications & Networks 4. Section 4: Media access methods: Space, time, frequency, & Code Division. 5. Section 5: Cellular Concept, System Design issues, satellites. 6. Section 6: GSM (Global System for Mobile Communications). 7. Section 7: Portable information appliances: laptops, notebooks, sub- notebooks, and MNCs hand-held computers PDAs and smartphones. 8. Section 8: Internet: TCP/ IP & de- facto application protocols.

	9. Section 9: Desktop Vs Mobile Device architectures. ARM architecture. What makes the mobile environment different? 10. Section 10: Software architecture for mobile computing. 11. Section 11: Mobile Operating Systems (Android and iOS) 12. Section 12: Security in Mobile Systems. 13. Section 13: Mobile browsers: WebKit.
Examination forms	Class discussion; mid - term and final - term exam: written exams
Study and examination requirements	Class attendance: 5%. Discussion, exercises, practices: 5%. Midterm exam: 20% Final theory exam: 70%. Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Jochen Schiller, *Mobile Communications*, 2nd edition, ISBN-13: 007-6092019329. References: 1. *Handbook of Wireless Networks and Mobile Computing* (2002) 2. *Principles of mobile computing and communications* by Othman, Mazliza (2008) 3. *Mobile computing* by Kamal, Raj (2007)

Advanced Artificial Intelligence - MTH10356

Module designation	Advanced Artificial Intelligence
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assoc. Prof. Nguyen Thanh Binh

Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course will present topics in Artificial Intelligence (AI). We will begin by defining the term "software agent" and discussing how software agents differ from programs in general. We will then look at those problems in the field of AI that tend to receive the most attention. Different researchers approach these problems differently.
Content	This module includes the following topics: Solving Problems by Searching Knowledge Representation and Reasoning
Examination forms	Final exam: written exams
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Peter, Norvig, and Russell Stuart *Artificial Intelligence. A Modern Approach.* Pearson Education, USA, 2021. References:

	1. Christian, Brian. *The alignment problem: How can machines learn human values?*. Atlantic Books, 2021. 2. Narayanan, Arvind, and Sayash Kapoor. *"AI snake oil: What artificial intelligence can do, what it can't, and how to tell the difference."* AI Snake Oil. Princeton University Press, 2024.

Natural Language Processing - MTH10359

Module designation	Natural Language Processing
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Huynh Thanh Son
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	lecture, lesson, lab work, project.
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Introduction to Machine Learning (MTH10353) ● Introduction to Artificial Intelligence (MTH10318)
Module objectives/intended learning outcomes	Equip students with the basic knowledge of natural language processing (NLP), the methods and techniques used in NLP, along with fundamental mathematical models applied to text processing tasks in non-random contexts.

Content	Including the theory of natural language processing (NLP) models, syntactic and semantic analysis, language generation, and the impact of computational methods and algorithms in NLP applications.
Examination forms	lab work and oral presentation
Study and examination requirements	lab works: 30%, oral presentation: 70%
Reading list	Main textbooks: 1. *Natural Language Processing in Action: Understanding, Analyzing, and Generating Text with Python* (2019) References: 1. *Practical Natural Language Processing* - O'Reilly (2020) 2. *Natural Language Processing with Transformers* - O'Reilly (2022) 3. *Transformers for Natural Language Processing: Build Innovative Deep Neural Network Architectures for NLP with Python, PyTorch, TensorFlow, BERT, RoBERTa, and More* (2021)

4. SPECIALIZATION IN APPLIED MATHEMATICS AND COMPUTER SCIENCE

4.1. COMPULSORY COURSES

Object-oriented software development - MTH10308

Module designation	Object-Oriented Software Development
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Pham Thi Vuong
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Lab Works: 30 hours (in class) Practice: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: To be familiar with object-oriented software development. Specific objectives/course learning outcomes: 1. Knowledge: Provide students with in-depth knowledge related to the main subjects in the field of object-oriented software engineering (technology processes, implementation techniques, tools, and deployment of environments software).

	2. Skills: research and work in the field of object-oriented software development. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 1. Basic concepts of object-oriented software development. 2. Object model, state model, functional model. 3. System design, object design, interface design. 4. Use IDEs and source code management tools.
Examination forms	Project
Study and examination requirements	1. Analytical and design skills (30%). 2. Skills in implementation, implementation, and testing (30%). 3. Document design skills (20%) 4. Presentation skills (20%) Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. *Object-Oriented Software Design in C++*, R. Mak, Manning Publications, 2024 References: 1. *Object Oriented Analysis and Design A Complete Guide*, G. Blokdyk, 5STARCooks, 2021 2. *Systems Analysis and Design: An Object-Oriented Approach with UML*, A. Dennis, B. H. Wixom, D. Tegarden, Wiley, 2020

System and Network Administration - MTH10309

Module designation	System and Network Administration
Semester(s) in which the module is taught	6th semester

Person responsible for the module	Dr. Vo Duc Cam Hai
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab work
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Lab Works: 30 hours (in class) Practice: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	4 Credits (7 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Computer Networks (MTH10311)
Module objectives/intended learning outcomes	General objectives: To be familiar with system and network administration. Specific objectives/course learning outcomes: 1. Knowledge: Provides knowledge about network services and principles of network administration in general. 2. Skills: Equip students with skills in installing and configuring network services on Unix/Linux server environments. After studying, students can take on a part of Unix/Linux server network administration in large companies or manage the entire Unix/Linux network for small and medium-sized businesses. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics:

	1. Chapter 1: In this module, students will have more in-depth knowledge of the Unix/Linux operating system environment. Students know how to partition and install a server using Unix/Linux operating systems to get the most out of it. In addition, students will learn about software package installation and management methods. Also, learn how to compile and optimize software features from source code. 2. Chapter 2: Focuses on user administration principles and practices. Administer system startup and shutdown mechanism. Administration of the file system and administration of some basic services. System log management. Set up an automatic data backup mechanism. 3. Chapter 3: Focus on configuring network parameters, learn some commands to look up network information, and then learn network security knowledge by firewall. 4. Chapter 4: Understanding the role of services on the internet. Conduct research and test software such as bind, vsftp, apache, postfix, courier, openssh. 5. Chapter 5: Learn the role of services on the intranet to set up a Workgroup network, a Domain Controller network to manage and share resources in the intranet. Manage access to internet resources. Conduct research and test software samba, dhcp, and squid. 6. Chapter 6: Introduction to directory services, configure settings, and prepare sample databases for directory services. Then proceed to integrate the services on the LDAP directory service.
Examination forms	Project; Midterm and final exam: written exams
Study and examination requirements	Diligence: 10% Midterm exam: 10% Project: 50%

	Final exam: 30% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. *Textbook on Fundamentals of System and Network Administration* (April 15, 2020) References: 2. *Mastering Linux Administration: Take your sysadmin skills to the next level by configuring and maintaining Linux systems, Alexandru* Calcatinge, Julian Balog, March 22, 2024 3. *Network and Computer Systems Administrators guide: Beginner's Guide to Networking and Computer Systems Administration for DevOps*, May 31, 2023

.Net Programming - MTH10310

Module designation	.NET Programming
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab work, practicing and discussing in groups
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)

Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Object Oriented Programming (MTH10407)
Module objectives/intended learning outcomes	Develop application programming skills with C# on the integrated programming environment Visual Studio .NET.
Content	Using the Visual Studio 2005 environment: design interfaces, perform application tasks, create software components... based on .NET, check for traps, handle errors, and deploy applications. This module includes the following topics: 1. Overview of .NET technology 2. Basic concepts in C# programming language 3. Windows Form 4. Actions with Files 5. Access databases with .NET 6. Deploy the application
Examination forms	Class discussion; midterm and final exam: written exam
Study and examination requirements	Class attendance: 5%. Discussion, exercises, practices: 5%. Midterm exam: 20% Final theory exam: 70%. Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. *Systems Programming with C# and .NET: Building robust system solutions with C# 12 and .NET 8*, D. Vroegop, Packt Publishing 2024. References: 1. *Pro C# 8 with .NET Core 3: Foundational Principles and Practices in Programming*, Andrew Troelsen, P. Japikse, Apress Berkeley 2020.

	2. *Pro C# 10 with .NET 6: Foundational Principles and Practices in Programming* (7th Ed.), A. Troelsen, P. Japikse, Apress 2022.

Computer Networks - MTH10311

Module designation	Computer Networks
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Vo Duc Cam Hai
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme. Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	Basic knowledge of computer networks such as distinguishing between different types of networks, understanding communication and data transmission bases in networks, and understanding the functions and services that operate at each layer of the OSI and TCP models. /IP. Students will learn skills in using software tools to capture and analyze data at each floor. Use popular services like web, file transfer, email, domain name, firewall. Configure the router device.
Content	This module includes the following topics: Chapter 1: Introduction to the overview of computer networks and the applications of computer networks in practice. Classify the types of computer networks and network operating systems. A brief introduction to the OSI and TCP/IP protocol stack. Chapters 2, 3, 4, 5: Introduce in detail the functions and services of the application, transport, network, and data link layers. The content taught in these chapters follows top-down approaches that will make it easy for students to acquire knowledge about networks. Chapter 6: Introduction to network security and basics such as encryption methods, authentication methods, understanding some types of network attacks and preventions to ensure network data integrity.
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Diligence: 10% Midterm exam: 10% Practice: 30% Final exam: 50% Final score is greater or equal to 5.0/10.0

Reading list	1. Fred Halsall, Computer Networking, and the Internet, Fifth edition, Addison Wesley, 2005 2. F Kurose, James. Computer networking: A top-down approach. Pearson Education, 2021 3. Andrew S. Tanenbaum, Nick Feamster, David J. Wetherall, Computer Networks, Pearson (6th Edition, 2021) 4. Douglas E. Comer, Computer Networks and Internets, Pearson (6th Edition, 2015)

Introduction to Database Systems - MTH10312

Module designation	Introduction to Database Systems
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc. Nguyen Hien Luong
Language	Vietnamese
Relation to curriculum	Compulsory for DS programme; Compulsory for Specialization in Data Science and Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	Introduce students to basic concepts of databases; data models; database design standards and standard forms.
Content	Basic concepts of databases. Linked entity data model, relational data model. Operations on the relational data model. SQL query language. Standard forms and normalization algorithms for databases.
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Mid-term: 30%, Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	1. R. Elsmari, S. Navathe. *Fundamentals of Database Systems 7th edition*, Addison Wesley, 2016. 2. Hector Garcia-Molina, Jeffrey Ullman, Jennifer Widom. *Database Systems: The Complete Book*. Pearson, 2008. 3. S. K. Singh. *Database Systems: Concepts, Design and Applications*. Pearson, 2009.

UNIX operating system - MTH10313

Module designation	UNIX Operating System
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Vo Duc Cam Hai
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture

Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Helps students have an overview of computer operating systems in general and in-depth knowledge of Unix/Linux operating systems in particular. Students will be equipped with skills in operating system installation, software package installation/compile, use of utilities and programming (shell, system) on Unix/Linux operating systems. The course will orient students to pursue a career in network system administration or software programming for embedded systems.
Content	1. Unix/Linux OS installation process. 2. Use Unix/Linux to serve end users: a. Unix/Linux external architecture overview. b. Basic utility programs in Unix/Linux. c. Install/compile software packages in Unix/Linux. d. Computer system administration. 3. Using Unix/Linux to serve shell programmers: a. Do the basics with Shells. Introduction of themes and corresponding usage scripts. b. Shell programming. Work with the Shells Script programming language and build application programs. 4. Using Unix/Linux to serve system programmers (C programmers):

	a. An overview of how to build and compile C programs in Unix/Linux. b. Overview of the internal architecture of the Unix/Linux OS (Unix/Linux Internal). 5. Introduce OS components such as file system, Process manager, and write demonstration programs to better understand OS working principle.
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Diligence: 10% Midterm exam: 10% Practice: 30% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. *Linux with Operating System Concepts* (Richard Fox, Publication Date :2021-12-29) References: 1. *Mastering Linux System Administration* (Publication Year: 2021) 2. *Linux for System Administrators: Navigate the Complex Landscape of the Linux OS and Command Line for Effective Administration* (Viorel Rudareanu, Daniil Baturin, Publication Year:2023)

Software Project Management - MTH10314

Module designation	Software Project Management
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Hien Luong
Language	Vietnamese

Relation to curriculum	Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	The document provides students at the first session including: summary lectures, lectures with slides, references. Students attend theoretical lectures and practice, participate in discussion, self-study, and major assignments.
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Object-oriented software development (MTH10308)
Module objectives/intended learning outcomes	Introduce students to the basics of planning, organizing, and managing software projects.
Content	1. Basic concepts of software project management. 2. GANTT, PERT chart. 3. Software standards, document standards. 4. Risks in the software. 5. Software configuration. 6. Software price estimation model.
Examination forms	Midterm exam: project or written. Final exam: written exam.
Study and examination requirements	Do theoretical and practical exercises in groups (30%). Projects, seminars (70%).

	Final score is greater or equal to 5.0/10.0
Reading list	1. P. Jalote. *Software Project Management in Practice,* Addison Wesley, 2002. 2. I. Somerville. *Software Engineering, Addison Wesley*, 1996. 3. WA Randolph. *Effective Project Planning and Management*, Prentice Hall, 1998.

Analysis and Design of Information Systems - MTH10315

Module designation	Analysis and Design of Information Systems
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Nguyen Hien Luong
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.
Teaching methods	Lecture, lab works, seminar, project
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Introduction to Database Systems (MTH10312) ● Object-oriented Programming (MTH10407)
Module objectives/intended learning outcomes	Introduce students to classical design and analysis methods used in functional analysis, data analysis, dynamic analysis, and systems design.

Content	1. Method of surveying the current situation and understanding the needs. 2. Structured analysis (SA). 3. Linked entity data model, relational data model. 4. SART method. 5. SD method.
Examination forms	Final exam: project
Study and examination requirements	Do theoretical and practical exercises in groups (30%). Projects, seminars (70%). Final score is greater or equal to 5.0/10.0
Reading list	1. Alan Dennis, Barbara Haley Wixom, Roberta M. Roth, *Systems Analysis and Design (8th Edition)*, Wiley, 2021. 2. Kenneth E. Kendall, Julie E. Kendall, *Systems Analysis and Design (11th Edition)*, Pearson, 2023. 3. Shouhong Wang, Hai Wang, *Information Systems Analysis and Design (2nd Edition)*, Universal Publishers, 2022. 4. Joseph S. Valacich & Joey F. George, *Modern Systems Analysis and Design (10th Edition)*, Pearson, 2024.

Java programming - MTH10316

Module designation	Java Programming
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Nguyen Tan Trung
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Applied Mathematics and Informatics, MCS programme. Elective for others programme.

Teaching methods	Lecture, lab work
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Object-Oriented Programming (MTH10407)
Module objectives/intended learning outcomes	Students will acquire basic advanced knowledge and skills in Java programming including basic programming, object-oriented programming, and interface programming. From there, students can learn more about J2EE technology such as programming applications for network and web environments, programming with databases, distributed programming, etc. and become a programmer on J2EE technology in the future.
Content	This module includes the following topics: 1. Overview of the Java language 2. Introducing the Java language 3. Objects and classes 4. Error management and garbage collection 5. I/O programming 6. Create user interface 7. Event modeling with AWT 8. Programming Menus with AWT 9. Multi-threaded programming
Examination forms	Midterm and final exam: written exams

Study and examination requirements	Attendance: 10% Midterm exam: 10% Practice: 30% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. *Headfirst Java*, 3rd Edition (2023) References: 1. *Clean code* (2008) 2. *Thinking in Java* (2006) 3. *Test-Driven Java Development* (2015)

4.2. ELECTIVE COURSES

Pattern Recognition - MTH10322

Module designation	Pattern Recognition
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture and Lab work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours, Lab work: 30 hours Private study: 90 hours (Self-study)
Credit points	4 Credits (6.5 ETCS)

Required and recommended prerequisites for joining the module	Required prerequisites: ● Computational Software Laboratory (MTH00087)
Module objectives/intended learning outcomes	This course provides fundamental knowledge of pattern recognition with examples from many applications, multidimensional data processing techniques, classification and regression algorithms as well as data clustering. Knowledge: familiarity with fundamental knowledge of pattern recognition such as multidimensional data processing techniques, classification algorithms, regression and data clustering Skills and competences: cognitive and practical abilities to use the knowledge in pattern recognition field, artificial neural network, SVM (support vector machine) and computer vision
Content	This module includes the following topics: 1. Basic concepts in pattern recognition 2. Bayesian discrimination principle 3. Parameter estimation 4. Non-parametric techniques 5. Linear discriminants 6. Support Vector Machine 7. Artificial Neural Network 8. Algorithm-independent machine learning 9. Unsupervised learning and clustering 10. Final Project
Examination forms	Class discussion; mid-term and final-term exam: written exam.
Study and examination requirements	Class attendance: 5%. Discussion, exercises, practices: 5%. Midterm exam: 20%

	Final theory exam: 70%. Final score is compulsory and greater or equal to 5.0/10.0.
Reading list	Main books: 1. *Fundamentals of Pattern Recognition and Machine Learning*, U.Braga-Neto, Springer 2020. References: 1. *Pattern Recognition: Introduction, Features, Classifiers and Principles*, J. Beyerer, R. Hagmanns, D. Stadler, De Gruyter 2024. 2. *Pattern Recognition, An Introduction*, S. T. Ahmed, S. M. Basha, S. R. Arumugam, M. M. Kodabagi, MileStone Research Publications 2021.

Seminar on Applied Mathematics and Computer Science - MTH10326

Module designation	Seminar on Applied Mathematics and Computer Science
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assigned lecturers of the Faculty of Mathematics and Computer Science, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Seminar; Project
Workload (incl. contact hours, self-study hours)	160 Hours Contact hours: Lectures: 45 hours (in class) Discussion: 30 hours (in class) Private study: 85 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	General objectives: Equipping students with basic knowledge to prepare for graduation thesis. Specific objectives/course learning outcomes: 1. Knowledge: Master the basic knowledge to prepare for graduation thesis. 2. Skills: basic skills to prepare for graduation thesis. 3. Attitude: have the right attitude, perspective and awareness about the subject.
Content	Students will work directly with faculty on a topic. Topic content can be: 5. Students must clearly understand their math problems 6. Find related documents. 7. What knowledge is related to this problem? The methods of our predecessors. 8. Synthesize this knowledge into documents. 9. Practice knowledge presentation skills within a certain period.
Examination forms	Class discussion; Final exam: project or representation.
Study and examination requirements	Class-attendance: 15% Discussion and practices: 35% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Depends on the assigned topics.

Image Transformation and Processing – MTH10331

Module designation	Image Transformation and Processing
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Pham The Bao

	Dr. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture and lab work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours, Lab work: 30 hours Private study: 90 hours (Self-study)
Credit points	4 Credits (6.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisite: ● Computer graphics (MTH10327) ● Data Structures & Algorithms (MTH10405) Recommended prerequisite: ● Computational Software Laboratory (MTH00087)
Module objectives/intended learning outcomes	This course provides basic knowledge of image processing, combining computer graphics and computer vision. Popular image transformations are used in visual effects or film effects. Specific objectives/course learning outcomes: 1. Knowledge: Mastering the basic knowledge of image transformation and processing. 2. Skills: Analyse and apply algorithms in practice. 3. Attitude: have the right attitude, perspective and awareness about the course.
Content	This module includes the following topics: 1. Overview of image 2. Common image and video processing methods 3. Image processing operations

	4. Camera model
Examination forms	Class discussion; mid-term and final exam: written exam.
Study and examination requirements	Class-attendance: 5% Discussion, exercises and practices: 5% Midterm exam: 20% Final theory exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, 2010. Reference: 1. Francis S Hill, *Computer graphics*, 1990. 2. Peter Shirley, Stephen Robert Marschner, *Fundamentals of computer graphics*, 2009. 3. *Digital Image Processing* by Rafael C. Gonzalez and Richard E. Woods (1992)

Software Testing - MTH10332

Module designation	Software Testing
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc. Nguyen Dat Thong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works

Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: To be familiar with software testing and software testing techniques. Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of software testing and software testing techniques, thereby understanding the importance of software testing in the software development process, understanding the role and work of testers. 2. Skills: research and work in the field of software testing. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 1. Overview of software testing 2. Basic concepts in software testing 3. Types of software testing 4. Software testing techniques 5. Project management in software testing
Examination forms	Class discussion; Midterm and final exam: written exams

Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. Hung Q. Nguyen, Bob Johnson, Michael Hackett and Robert Johnson, *Testing Web Applications*, VNUHCM Press, 2010. References: 1. Cem Kaner, Jack Falk and Hung Q. Nguyen, *Testing Computer Software*, 2nd Edition, Wiley, 1999. 2. Bernard Homès. *Fundamentals of Software Testing*. Wiley-ISTE, 2012.

Design and Build Web Applications - MTH10333

Module designation	Design and Build Web Applications
Semester(s) in which the module is taught	7th semester
Person responsible for the module	M.S. Vo Duc Cam Hai M.S. Tran Anh Tuan
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class) Private study: 75 hours (self-study)

Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Fundamentals of Computer Programming (MTH00055)
Module objectives/intended learning outcomes	General objectives: The course provides basic knowledge in the field of website design. Apply graphic design tools for websites. Specific objectives/course learning outcomes: 1. Knowledge: Master web application architecture, website design process, website building tools and languages. 2. Skills: survey, analysis, design, interface building and website deployment. 3. Attitude: Attend all classroom sessions, and discussions outside of class time; have the right attitude, perspective and awareness about the subject.
Content	This module includes the following topics: 1. Basic HTML 2. Advanced HTML 3. Basic JavaScript 4. Graphic design tools 5. Project
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Midterm exam: 40% Final exam: 60% Final score is greater or equal to 5.0/10.0

Reading list	Main textbooks: 1. *Front-End Web Development with Modern HTML, CSS, and JavaScript*, Packt Publishing, 2023 2. *Mastering Web Development: A Journey through HTML, CSS, and JavaScript*, independently published, 2023 3. *Advanced Web Application Architecture*, Matthias Noback, Leanpub, 2021 References: 1. Open-source web design documents, http://www.w3schools.com 2. MDN Web Docs (developer.mozilla.org)

Web Programming with J2EE - MTH10334

Module designation	Web Programming with J2EE
Semester(s) in which the module is taught	6th semester
Person responsible for the module	M.Sc. Ha Van Thao Dr. Vo Duc Cam Hai
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works.
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Lab Works: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	● Introduction to Database Systems (MTH10312) ● Java Programming (MTH10316)

Module objectives/intended learning outcomes	General objectives: Provide students with the knowledge to develop web applications using Java Enterprise Edition: 1. Servlet, JSP to build web applications 2. Concepts of Cookie, Session 3. MVC framework 4. JDBC, EL, … Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of building web applications. 2. Skills: survey, analyse, design, build, deploy and maintain websites. 3. Attitude: Attend all classroom sessions, and discussions outside of class time; have the right attitude, perspective and awareness about the subject.
Content	This module includes the following topics: 1. Servlet Technology 2. Form Processing 3. Header Processing 4. Cookie 5. Session Tracking 6. JSP Technology 7. Java Bean 8. MVC Framework 9. Expression Language
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Midterm exam: 50% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks:

	1. *Beginning Jakarta EE Web Development: Using JSP, JSF, MySQL, and Apache Tomcat for Building Java Web Applications 3rd ed.* (Publication date,August 5, 2020) References: 1. *Jakarta EE for Java Developers: Build Cloud-Native and Enterprise Applications Using a High-Performance Enterprise Java Platform* (English Edition, Publication date - December 16, 2021) 2. *Jakarta EE Application Development: Build enterprise applications with Jakarta CDI, RESTful web services, JSON Binding, persistence, and security,* Second Edition (Feb 2024)

Software Testing 2 - MTH10336

Module designation	Software Testing 2
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Nguyen Dat Thong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 30 hours (in class) Lab Works: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	3 Credits (5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Software Testing (MTH10332)

	● Introduction to programming (MTH00055) ● Object Oriented Programming (MTH10407)
Module objectives/intended learning outcomes	General objectives: To be familiar with software test automation and techniques in automated software testing. Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of software test automation and techniques in automated software testing, thereby understanding the importance of automated software testing in the software development process; Understand the role and work of automation testing engineers. 2. Skills: research and work in the field of software testing automation. 3. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 1. Overview of automated software testing 2. Automated unit testing with xUnit 3. Automated testing with Microsoft CodedUI 4. Automate testing of Web applications with Selenium 5. Automated performance testing with Microsoft Visual Studio
Examination forms	Class discussion; Final exam: written exams
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final practice exam: 50% Final theory exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks:

	1. Hung Q. Nguyen, Bob Johnson, Michael Hackett and Robert Johnson, *Testing Web Applications*, VNUHCM Press, 2010. References: 1. Cem Kaner, Jack Falk and Hung Q. Nguyen, *Testing Computer Software*, 2nd Edition, Wiley, 1999. 2. Selenium HQ: http://docs.seleniumhq.org/docs/ 3. JUnit: http://junit.org/junit4/ 4. NUnit: https://www.nunit.org/index.php?p=documentation 5. CodedUI Test: https://msdn.microsoft.com/en-us/library/dd286726.aspx

Web Programming with PHP - MTH10337

Module designation	Web Programming with PHP
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Tran Anh Tuan M.S. Nguyen Dat Thong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture; Lab Works.
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Lab Works: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites:

	• Introduction to Database Systems (MTH10312)
Module objectives/intended learning outcomes	General objectives: Provide students with the knowledge to build and develop web applications using PHP technology. Specific objectives/course learning outcomes: 1. Knowledge: Mastering the knowledge of building web applications. 2. Skills: survey, analyze, design, build, deploy and maintain websites. 3. Attitude: Attend all classroom sessions, and discussions outside of class time; have the right attitude, perspective and awareness about the subject.
Content	This module includes the following topics: 1. Introduction 2. Basic PHP 3. Advanced PHP 4. PHP and databases 5. PHP and XML 6. Deploying web applications 7. Introduction to PHP frameworks 8. Web security issues
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Midterm exam: 40% Final exam: 60% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks: 1. *PHP for the Web – Learning PHP without a framework* (Matthias Noback, 2022) References:

	1. *Front Line PHP: Building modern applications with PHP 8* (Brent Roose, Roman Pronskiy (Foreword), February 9, 2021) 2. *Learning PHP, MySQL & JavaScript: A Step-by-Step Guide to Creating Dynamic Websites* (Robin Nixon, August 31, 2021)

Pattern Recognition and Analysis - MTH10340

Module designation	Pattern Recognition and Analysis
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Pham The Bao Dr. Huynh Trung Hieu
Language	Vietnamese
Relation to curriculum	Elective ● Compulsory for Specialization Mathematical Methods in Computer Science, MCS programme. ● Elective for others programme.
Teaching methods	Lecture; Lab Works.
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Lab Works: 30 hours (in class) Practice: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Introduction to Artificial Intelligence (MTH10318)
Module objectives/intended learning outcomes	General objectives: Introduce the mathematical foundations of pattern recognition with examples from a variety of

	applications. Multidimensional data processing techniques, classification and regression algorithms as well as data clustering will be presented. Students and graduate students interested in pattern recognition, artificial neural networks, SVM (support vector machine) and computer vision can join this course. Specific objectives/course learning outcomes: 1. Knowledge: Master the basic knowledge of image transformation and processing. 2. Skills: analyze and apply algorithms in practice. 3. Attitude: Attend all classroom sessions, and discussions outside of class time; have the right attitude, perspective and awareness about the subject.
Content	This module includes the following topics: 1. Introduction 2. Bayesian discrimination principle. 3. Parameter estimation 4. Non-parametric techniques 5. Linear discriminants 6. Artificial Neural Network 7. Support Vector Machine 8. Algorithm independence in machine learning 9. Unsupervised learning and clustering
Examination forms	Class discussion; Midterm and final exam: written exams
Study and examination requirements	Class attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	Main textbooks:

	1. *Pattern Recognition and Data Analysis with Applications*, D. Gupta, R. S. Goswami, S. Banerjee, M. Tanveer, R. B. Pachori, Springer Nature, 2022. References: 1. *Fundamentals of Pattern Recognition and Machine Learning*, U. M. Braga-Neto, Springer, 2021 2. *Analytic Learning Methods for Pattern Recognition*, K.A. Toh, H. Zhuang, S. Liu, Z. Lin, Springer, 2025

Database Management System - MTH10344

Module designation	Database Management System
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Tran Anh Tuan MSc. Nguyen Hien Luong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lab works
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Introduction to Database Systems (MTH10312)
Module objectives/intended learning outcomes	Students know how to access knowledge about databases and database management systems: components of the SQL Server database management system and their functions,

	concurrent access management mechanisms, safety and data recovery after incidents, decentralisation and security.
Content	Introduction including definition, functions, properties, architecture of database management system, SQL and Microsoft SQL server Fundamental SQL syntaxes Securing SQL server Stored and manage procedures Creating and managing user-defined functions Creating and managing triggers in SQL server
Examination forms	Midterm and final exam: written exam
Study and examination requirements	Class-attendance: 5% Practices and exercises: 5% Midterm exam: 20% Final exam: 70% Final score is greater or equal to 5.0/10.0
Reading list	1. Duong Quang Thien. *SQL Server 2000: T-SQL Programming.* Ho Chi Minh City: Saigon Culture Publishing House, 2007. 2. Ray Rankins, Paul Bertucci, Chris Gallelli, Alex T. Silverstein, *Microsoft SQL Server 2005 Unleashed*, Sams Publishing, 2007. 3. Brian Knightet al, *Professional SQL Server 2005 Administration*, Wrox Press, 2007. 4. Paul Turley & Dan Wood, *Beginning Transact-SQL with SQL Server 2000 and 2005*, Wrox Press, 2006.

Natural Language Processing - MTH10359

Module designation	Natural Language Processing
Semester(s) in which the module is taught	6th semester

Person responsible for the module	MSc. Huynh Thanh Son
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	lecture, lesson, lab work, project.
Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: ● Introduction to Machine Learning (MTH10353) ● Introduction to Artificial Intelligence (MTH10318)
Module objectives/intended learning outcomes	Equip students with the basic knowledge of natural language processing (NLP), the methods and techniques used in NLP, along with fundamental mathematical models applied to text processing tasks in non-random contexts.
Content	Including the theory of natural language processing (NLP) models, syntactic and semantic analysis, language generation, and the impact of computational methods and algorithms in NLP applications.
Examination forms	lab work and oral presentation
Study and examination requirements	lab works: 30%, oral presentation: 70%
Reading list	Main textbooks: 1. *Natural Language Processing in Action: Understanding, Analyzing, and Generating Text with Python* (2019) References:

	1. *Practical Natural Language Processing* — O'Reilly (2020) 2. *Natural Language Processing with Transformers* — O'Reilly (2022) 3. *Transformers for Natural Language Processing: Build Innovative Deep Neural Network Architectures for NLP with Python, PyTorch, TensorFlow, BERT, RoBERTa, and More* (2021)

5. GRADUATING WORKS

Internship - MTH10549

Module designation	Internship
Semester(s) in which the module is taught	8th semester
Person responsible for the module	Assigned lecturers of the Faculty of Mathematics and Computer Science, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Instructions, hands-on activities
Workload (incl. contact hours, self-study hours)	Total workload: 180 hours Contact hours (lectures, exercises): 90 hours Private study including examination preparation, specified in hours: 90 hours
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: 1. Apply knowledge, skills in mathematics/applied mathematics/computer science to practice with real-life problems in company, industry. 2. Self-orient in the desired professional field. 3. Understand the impact of mathematics and computer science in a global, economic, environmental, and societal context. 4. Conduct work responsibly in cooperation with others. 5. Write and present a report in adequate format.
Content	1. Attendance: Students should attend 100%. Attendance will be regulated and checked. 2. Individual Assignments: Tasks are assigned by the industrial company. 3. Internship: At a company in the field corresponding to the registered major.
Examination forms	Internship report, Oral presentation
Study and examination requirements	Internship certificate issued by the company: 50% Written report and presentation graded by the faculty supervisor: 50%
Reading list	Depending on the assigned tasks and instructors.

Graduation Thesis - MTH10595

Module designation	Graduation Thesis

Semester(s) in which the module is taught	8th semester
Person responsible for the module	Assigned lecturers of the Faculty of Mathematics and Computer Science, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Project, seminar
Workload (incl. contact hours, self-study hours)	Total workload: 300 hours Contact hours: 30 hours Private study including examination preparation, specified in hours: 270 hours
Credit points	10 credits (20 ECTS)
Required and recommended prerequisites for joining the module	Successfully completed 96/131 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: 1. Point out the research problems 2. Apply fundamental knowledge and theories to analyse the research problem; develop a research framework (model if applicable); and design a solution for the problem 3. Use appropriate techniques for solutions 4. Demonstrate the scientific contribution and practical relevance of the research carried out 5. Provide a consistent, well-structured Bachelor Thesis 6. Plan and manage learning process 7. Present and defend results of the thesis at the Students' Bachelor Thesis conference.
Content	Various topics
Examination forms	Graduation report
Study and examination requirements	Final defense: 100%
Reading list	Diverse and depends on the supervisors

Surveys of Mathematics, Computer Science, and specializations - MTH10617

Module designation	Surveys of Mathematics, Computer Science, and specializations.
Semester(s) in which the module is taught	4th semester

Person responsible for the module	The representative of each major.
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lectures, group work
Workload (incl. contact hours, self-study hours)	Total workload: 90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General Objective: The course aims to provide students with a broad understanding of mathematics, computer science, and related majors in the training program. This foundation helps students orient themselves and make informed decisions when selecting their major. Specific Objectives / Course Learning Outcomes: Knowledge: Students will gain an overview of mathematics, computer science, and related majors, along with their applications and career opportunities. Skills: Gaining a general understanding of a field. Attitude and Diligence: Seriousness, diligence, and a scientific approach.
Content	The course consists of lectures and activities conducted by multiple instructors, providing an overview of mathematics and computer science, their history worldwide, and the current state in Vietnam. It also introduces various majors, including their focus, research scope, applications, and career opportunities. Students engage actively, provide feedback, and undergo assessments.

Examination forms	written exams
Study and examination requirements	Tests - process score: 70% Final exam: 30 %
Reading list	1. *History of Mathematics* (in Vietnamese), Nguyen Phu Loc, Vietnam Education Publishing House, 2008. 2. *The History of the Development of Information Technology and its Organizational and Societal Impact*. M.C. Mr, 2013. 3. Lecture notes.